

Fast and Private Max-Sum Diversification (Full Version)

Ron Zadacario
Tel Aviv University
ronzadacario@mail.tau.ac.il

Tova Milo
Tel Aviv University
milo@post.tau.ac.il

ABSTRACT

Result diversification is crucial for generating informative, non-redundant data summaries and query outputs. Although its various formulations have been extensively studied across an array of data-driven disciplines, existing methods fail to address the privacy concerns that arise when the underlying data is sensitive. In this work, we initiate the study of result diversification under *differential privacy*, focusing on the *max-sum diversification* (MSD) problem, a widely adopted model with the objective of maximizing a linear combination of a submodular function, quantifying relevance, and the sum of pairwise distances between selected items, quantifying diversity. We propose differentially private algorithms for MSD under both cardinality and matroid constraints, achieving nearly optimal utility guarantees. At the same time, we design more efficient algorithms that maintain strong guarantees. Notably, the proposed algorithms are faster than existing non-private methods, making them appealing even in non-private settings. Experimental evaluations on real-world datasets demonstrate that the proposed approach achieves utility comparable to that of non-private baselines even under strong privacy guarantees, and significantly improves execution times for cardinality constraints.

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The source code, data, and/or other artifacts have been made available at <https://github.com/ronzadi/Differentially-Private-Max-Sum-Diversification>.

1 INTRODUCTION

Result diversification is the problem of selecting a subset of items that is both relevant and diverse, subject to feasibility constraints. It underlies numerous data summarization and selection tasks, where capturing data variety, avoiding redundancy, and providing informative outputs are crucial. Consequently, its various formulations have been extensively studied across multiple domains, including databases, machine learning, and operations research. Among these formulations, the *max-sum diversification* (MSD) problem is one of the most well-studied and widely adopted diversity models in the

database literature. It has been applied to a range of tasks, including query result diversification [3, 8, 20, 29, 56], graph summarization [32, 54], rule mining [7, 34], spatial keyword search [62], as well as recommender systems [11, 18, 48] and feature selection [30, 59]. In this problem, we are given a set of items, referred to as the *ground set*, together with pairwise distances and a monotone submodular set function f that quantifies the *relevance* of each subset. The objective is to select a subset S that maximizes a linear combination of $f(S)$ and the sum of pairwise distances among the items in S , subject to given feasibility constraints [8]. Prior work has primarily focused on cardinality constraints, which limit the size of the selected subset, and more generally on *matroid* constraints (defined in Section 3), which capture a broad range of practical feasibility criteria. A notable special case of this model is that of diverse Top- k queries [31], where each item v is associated with a score $q(v)$, the relevance of a subset S is given by $f(S) = \sum_{v \in S} q(v)$, and the constraint is cardinality $|S| \leq k$. By allowing arbitrary monotone submodular relevance functions and matroid constraints, MSD supports a substantially more expressive class of data summarization tasks, as illustrated in Example 1.1.

Yet, many compelling applications of result diversification, and of MSD in particular, involve sensitive individual data, such as purchase histories, locations, and search patterns. Improper handling of such data can pose significant privacy risks [57, 61], and therefore the maximization procedure must provide formal privacy guarantees for the individuals contributing to the dataset.

Example 1.1. An online retailer wishes to select a set S of k popular products from a dataset filtered by category = “Health Care”, for purposes such as catalog design, promotion, or trend analysis. Selecting the Top- k most purchased items may primarily reflect the preferences of a narrow group of highly active users. To instead maximize user reach, the retailer can maximize the coverage function $f(S) = \frac{1}{m} |\bigcup_{v \in S} \mathcal{U}(v)|$, where $\mathcal{U}(v)$ denotes the set of users who purchased item v and m is the total number of users. This objective is monotone submodular and often used for summarization [19]. However, optimizing coverage alone may still yield a redundant set of items: Figure 1a shows results on the Amazon Reviews dataset [37], where three of the six selected items (indicated by shaded rows) are essential oils. Selecting overly similar products may neglect users’ varied interests and reduce long-term satisfaction [12].

To encourage diversity, each product is associated with a set of subcategories, and pairwise distances are defined using the Jaccard distance [39] between these sets. The retailer can then maximize a linear combination of $f(S)$ and the sum of pairwise distances among items in S , producing a relevant and diverse set of products. Additional constraints can be incorporated naturally. For example,

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Product	Subcategory	Product	Subcategory
Muscle Cream	Treatments	Muscle Cream	Treatments
Cassia Oil	Essential Oils	Cassia Oil	Essential Oils
Oil Diffuser	Diffusers	Oil Diffuser	Diffusers
Heating Pad	Hot & Cold Therapies	Heating Pad	Hot & Cold Therapies
Garlic Oil	Essential Oils	Shoe Insoles	Foot Health
Rosemary Oil	Essential Oils	Ear Otoscope	Ear Care
Coverage: 0.15	Diversity: 0.20	Coverage: 0.145	Diversity: 0.31

(a) **Relevance-only selection.** Three of six items are from the “Essential Oils” subcategory (shaded). (b) **Selection balancing relevance and diversity.** Shaded items are newly added.

Figure 1: Representative products from the Amazon Health Care category. Product and subcategory names are simplified for readability.

limiting the number of selected products within each price category induces a matroid constraint.

Critically, the selection procedure must also preserve users’ privacy. Since the objective is derived from individual transaction logs, an exact solution could inadvertently reveal sensitive information. For instance, the inclusion of a niche product could signal the presence of a user with a rare medical condition. Preventing such leakage necessitates a mechanism that provides formal privacy guarantees. Using the algorithms proposed in this work, the retailer can generate a relevant, diverse, and privacy-preserving summary, as illustrated in Figure 1b, where the shaded rows indicate two products added by the private and diversity-aware algorithm.

Differential Privacy [22, 24] is a rigorous notion that has become the gold standard for analyzing sensitive data under strong privacy guarantees, and has been adopted by multiple companies [2, 23] and governmental organizations [21, 26, 55]. Intuitively, the output distribution of a differentially private (DP) algorithm changes only minimally when a single individual’s data is modified, thereby ensuring that individual-level information is not disclosed (see Section 3 for the formal definition). Privacy is typically achieved by injecting noise into the computation process, which may result in some degradation in utility. In our setting, achieving stronger privacy often means computing a set of items with a slightly lower quality score. Although DP submodular maximization has received considerable attention in recent years [16, 33, 43, 49], the design of DP algorithms for the more general MSD problem has yet to be addressed. Existing techniques for DP submodular maximization do not directly apply to MSD, as its objective function is not submodular.

In this work, we study the MSD problem under differential privacy. We propose DP algorithms that provide near-optimal utility guarantees under both cardinality and general matroid constraints. Our proposed algorithms also improve time complexity compared to existing non-private methods, making them valuable even in non-private settings. Experimental evaluation on two real-world applications, Amazon product and Uber pickup location summarization, shows that our approach achieves utility comparable to non-private baselines while significantly reducing the number of

objective evaluations. We now state our contributions, with a summary of the main results in Table 1. To provide context, Section 2 reviews relevant lower bounds from prior work.

1.1 Our Results

In the DP setting, the MSD objective function depends on a sensitive dataset D and is a linear combination of a submodular relevance term and a diversity term. It is called *decomposable* if it can be expressed as a sum of functions, each depending on a single record of D . The formal definitions are provided in Section 4.

We begin with a direct DP adaptation of the greedy algorithm of Borodin et al. [8], which serves as a natural baseline. Building on this, our main contribution is a faster DP algorithm whose analysis goes well beyond the original work, simultaneously addressing subsampling, privacy, and the absence of submodularity.

Greedy algorithm for cardinality constraints. Cardinality constraints, which capture the fundamental Top- k query class, are among the most commonly studied constraint types. Borodin et al. [8] proposed a *non-oblivious*¹ greedy algorithm for cardinality-constrained MSD, achieving a tight $1/2$ -approximation. We extend their analysis to account for the error introduced by replacing greedy selections with privatized ones. For decomposable objectives, we show that the technique of Gupta et al. [33] provides improved privacy guarantees. The precise results, corresponding to row (i) of Table 1, are discussed in Section 5.

However, this algorithm makes $\Theta(nk)$ value oracle calls², where k is the cardinality bound and n is the ground set size. For large k (e.g., $k = \Omega(n)$), this complexity reaches $\Omega(n^2)$ oracle calls. Such scaling becomes a significant bottleneck in high-concurrency environments or interactive systems where each execution of the algorithm corresponds to a single user query and many such queries must be processed simultaneously. Even in well-resourced systems, the cumulative computational overhead of $\Theta(nk)$ evaluations per instance limits total system throughput and increases latency, underscoring the need for more efficient algorithms that better suit high-volume workloads.

Faster greedy algorithm via subset sampling. Algorithms for submodular maximization are often accelerated by “sample greedy” methods, where the next element is chosen from a smaller random subset rather than the entire ground set, improving efficiency at the cost of a slightly weaker approximation guarantee [10, 42, 43]. While these analyses rely on submodularity, which does not hold in our setting, we nonetheless propose an efficient “sample greedy” algorithm for the DP MSD problem with cardinality constraints. To our knowledge, this is the fastest known algorithm for MSD, making it valuable even in non-private settings.

We present two instantiations of the algorithm, each illustrating a distinct trade-off between complexity and utility. Furthermore, we extend the technique of Gupta et al. [33] to incorporate the subsampling step and prove stronger guarantees for decomposable objectives. Following established literature in both private [43] and non-private [10, 42] submodular maximization, our utility guarantees are provided in expectation, as the randomness inherent in the

¹An optimization algorithm is non-oblivious if it selects the next element with respect to an auxiliary function rather than the true objective.

²Number of evaluations of the objective function; see Section 3.

Table 1: Expected utility guarantees of (ϵ, δ) -DP algorithms and their query complexity. Here Δ is the sensitivity (equivalent to the decomposability parameter for decomposable objectives), n is the ground set size, k is the maximal feasible solution size, and γ controls the utility-complexity trade-off. For general Δ -sensitive functions with cardinality constraints, additive errors incur an additional $\sqrt{k/\log(1/\delta)}$ factor. Setting $\epsilon = \infty$ yields non-private algorithms with the same multiplicative approximation ratios. The $1/2$ -approximation ratio is tight under standard complexity-theoretic assumptions (see Section 2).

	Constraint	Approx.	Additive Error (Δ -decomposable)	Oracle Calls	Algorithm	
R2-W2	k-Cardinality	1/2	$O(\Delta k \epsilon^{-1} \log \delta^{-1} \log n)$	$O(nk)$	DP-Greedy (Algorithm 1)	(i)
R2-W2		$1/2 - \gamma$	$O(\Delta k \epsilon^{-1} \log \delta^{-1} \log n)$	$O(n \log k \log \gamma^{-1})$	DP-NOSG (Algorithm 2)	(ii)
R2-W2		$1 - 2/e - \gamma - 1/k$	$O(\Delta k \epsilon^{-1} \log \delta^{-1} \log n)$	$O(n \log \gamma^{-1})$	DP-OSG (Algorithm 2)	(iii)
R2-W2	Matroid (rank k)	$1/2 - \gamma$	$O\left(\frac{\Delta \sqrt{k \log k \log \delta^{-1}}}{\epsilon \sqrt{\gamma}} (k \log n + \log \gamma^{-1})\right)$	$O(\gamma^{-1} n k \log k)$	DP-SLS (Algorithm 3)	(iv)

subsampling process typically precludes the strong concentration required for meaningful high probability bounds. The precise results, corresponding to rows (ii) and (iii) of Table 1, are detailed in Section 6.

Local search for matroid constraints. Some data summarization tasks require more general constraints than simple cardinality, and matroids are among the most studied such constraints. Since the greedy paradigm does not provide any constant-factor approximation for the MSD problem under matroid constraints, Borodin et al. [8] showed that a natural single-swap local search algorithm achieves a $(1/2 - \gamma)$ -approximation. However, checking for local optimality or ensuring the selection of improving swaps is infeasible under differential privacy. To address this, we propose a DP local search algorithm for MSD under matroid constraints that performs a predefined number of local search steps and selects swaps privately. The analysis bounds the loss incurred by allowing non-improving swaps, yielding near-optimal guarantees. We further improve efficiency by sampling a subset of candidate swaps in each iteration, reducing to $O(\gamma^{-1} k \log k)$ oracle calls, a substantial improvement over the $O(n^2 + \gamma^{-1} n k^2 \log k)$ by Borodin et al. [8]. *To our knowledge, this algorithm achieves the best known complexity for MSD under matroid constraints.*

Importantly, even without privacy, the algorithm achieves (in expectation) the same approximation ratio as Borodin et al. [8] with improved complexity. The precise statements of our results, corresponding to row (iv) of Table 1, are given in Section 7.

2 RELATED WORK

We provide a brief survey of relevant literature, starting with non-private settings, followed by lower bounds, and finally covering additional privacy-related contributions.

Non-private result diversification. The MSD problem generalizes the *max-sum dispersion* problem (also called *remote-clique*), which aims to maximize $\sum_{\{u,v\} \subseteq S} d(u,v)$ and has attracted sustained attention for over three decades [27, 35, 36, 50]. Gollapudi and Sharma [31] first studied MSD for modular relevance and cardinality constraints, obtaining a $1/2$ -approximation by reduction to max-sum dispersion, which was later extended to matroid constraints by Abbassi et al. [1]. Borodin et al. [8] substantially generalized this framework to submodular relevance and matroid constraints,

providing a greedy algorithm for cardinality constraints and a local search algorithm for general matroid constraints with $1/2$ and $(1/2 - \gamma)$ approximations, respectively. For Euclidean distances with modular relevance, Indyk et al. [38] introduced composable coresets for streaming and distributed settings. Ceccarello et al. [14] later generalized these constructions to metrics with bounded doubling dimension, and Ceccarello et al. [13] extended the framework to matroid constraints. A fully dynamic variant for doubling metrics was studied by Pellizzoni et al. [47]. Agarwal et al. [4] proposed near-linear-time algorithms using efficient indexes for Euclidean distances. Additionally, Cevallos et al. [15] studied MSD under negative-type distances, a setting that is incomparable to general metric distances. Despite the extensive literature, none of these works provides formal privacy guarantees.

DP submodular maximization. As the MSD problem has not been studied in the context of differential privacy, we review the most closely related line of work, which focuses on DP submodular maximization. The work of Gupta et al. [33] was the first to address the problem, focusing on decomposable objective functions under cardinality constraints. Mitrovic et al. [43] considered the more general case of submodular objectives with bounded sensitivity, proposing algorithms for both monotone and non-monotone functions under matroid and p -extendable systems constraints. Chaturvedi et al. [16] revisited decomposable objectives, extending the results of Gupta et al. [33] from cardinality to matroid constraints and from monotone to non-monotone functions, improving upon Mitrovic et al. [43]. Rafiey and Yoshida [49] studied private submodular and k -submodular maximization under matroid constraints, achieving improved running time at the cost of larger additive error. Other notable advances include Salazar and Cummings [52], who addressed the online setting under cardinality constraints, Sadeghi and Fazel [51], who studied bounded-curvature objectives under matroid and knapsack constraints in both offline and online settings, and Chaturvedi et al. [17], who proposed a streaming algorithm for cardinality constraints. Yet, all of these approaches apply only to submodular objectives and do not directly extend to MSD.

Hardness and lower bounds. Cardinality-constrained max-sum dispersion is known to be NP-hard even with metric distances [35]; hence, the more general MSD problem is also NP-hard.

Moreover, Borodin et al. [8] showed that achieving an approximation factor better than $1/2$ is hard under the planted-clique hardness assumption [5]. As summarized in Table 1, the guarantees in rows (i), (ii), and (iv) match or nearly match this tight approximation factor. In contrast, row (iii) achieves improved complexity while providing a slightly weaker approximation guarantee.

Additionally, the DP MSD problem in Section 4 generalizes DP monotone submodular maximization, and hence known lower bounds for that problem extend to our setting. In particular, Gupta et al. [33] proved that any ε -DP algorithm for maximizing a submodular function under a cardinality constraint k over a ground set of size n must incur expected additive error $\Omega(\Delta k \varepsilon^{-1} \log(n/k))$. Chaturvedi et al. [17] extended this to the (ε, δ) -DP setting, showing that any such algorithm achieving multiplicative factor c must incur additive error $\Omega(\Delta k \varepsilon^{-1} c \log(\varepsilon/\delta))$, assuming $n \geq \delta^{-1} k (e^\varepsilon - 1)$ and $c \geq 4\delta/(e^\varepsilon - 1)$. In both cases, the hard instances are 1-decomposable, implying that our results in rows (i)–(iii) of Table 1 match the corresponding lower bounds up to logarithmic factors. These results recover the best-known additive error bounds for submodular maximization, and note that even in this restricted setting the tightness of logarithmic factors remains open [33]. For general bounded-sensitivity functions (row (iv), and rows (i)–(iii) with an additional $\sqrt{k/\log(1/\delta)}$ factor), our guarantees incur an additional $\sqrt{k}$ -factor compared to the known lower bounds, which are not known to be tight in this more general setting. A similar $\tilde{O}(\sqrt{k})$ gap also appears in the state-of-the-art results for bounded-sensitivity submodular maximization under cardinality constraints [43].

3 PRELIMINARIES

In this section, we present the basic definitions and notation, introduce submodular set functions and matroids, and review key concepts from differential privacy relevant to this work.

Notation and Set Functions. Let V denote a finite ground set of size n . For a set function $f : 2^V \rightarrow \mathbb{R}$, the *marginal contribution* of an element $u \in V$ to a set $S \subseteq V$ is $f(u \mid S) = f(S \cup \{u\}) - f(S)$. A function $d : V \times V \rightarrow \mathbb{R}_{\geq 0}$ is a *pseudometric* if it is symmetric, satisfies the triangle inequality, and $d(u, u) = 0$ for all $u \in V$. Extend d to sets via $d(S) = \sum_{\{u,v\} \subseteq S} d(u, v)$ and $d(S, T) = \sum_{u \in S, v \in T} d(u, v)$ for disjoint sets $S, T \subseteq V$. For $u \in V$, write $d(u \mid S) = d(S \cup \{u\}) - d(S) = \sum_{v \in S} d(u, v)$.

Submodular Functions and Matroids. A set function $f : 2^V \rightarrow \mathbb{R}$ is *monotone* if $f(S) \leq f(T)$ for all $S \subseteq T$, *non-negative* if $f(S) \geq 0$ for all $S \subseteq V$, and *submodular* [46] if $f(u \mid S) \geq f(u \mid T)$ for all $S \subseteq T$ and $u \in V \setminus T$. A *matroid* [58] is a pair $(V, \mathcal{I})$, where $\mathcal{I} \subseteq 2^V$ satisfies (i) if $A \subseteq B$ and $B \in \mathcal{I}$ then $A \in \mathcal{I}$, and (ii) for $A, B \in \mathcal{I}$ with $|A| < |B|$, there exists $u \in B \setminus A$ such that $A \cup \{u\} \in \mathcal{I}$. Subsets in $\mathcal{I}$ are called *independent sets*, and inclusion-wise maximal independent sets are called *bases*. All bases have the same cardinality, known as the *rank* of the matroid. See Section A for additional background and examples of matroids.

Complexity Model. We adopt the oracle complexity model, the standard abstraction in set function optimization, as it captures algorithmic complexity independently of application-specific details such as the cost of evaluating the objective function. Thus, we assume access to f , d , and $\mathcal{I}$ via oracles: a *value oracle* returns

$f(S)$ or $d(S)$ for a given S , and an *independence oracle* determines whether $S \in \mathcal{I}$. Complexity is measured by the number of oracle calls. While each marginal gain $d(u \mid S)$ may require up to k pairwise evaluations $d(u, v)$, we treat d as a set function and count each evaluation of $d(S)$ as a single oracle call for consistency with the submodular objective. Although the number of oracle calls is generally a reliable proxy for runtime, discrepancies may arise in practice, as evaluating the oracle on larger sets can be slower in some applications.

Differential Privacy. A dataset D is a finite multiset of records from a domain $\mathcal{X}$. The number of records in D is denoted by $m = |D|$. We let $\mathcal{D}$ denote the space of all possible datasets over this domain. Two datasets D and D' are called *neighboring* (denoted $D \sim D'$) if they differ in one record. Intuitively, differential privacy ensures that the distribution of outputs of a randomized algorithm does not significantly change when the data of a single individual is changed.

Definition 3.1 (Differential Privacy [22, 24]). A randomized algorithm $\mathcal{A}$ is (ε, δ) -differentially private (DP) if for any neighboring datasets $D \sim D'$ and any set of possible outputs $S \subseteq \text{Range}(\mathcal{A})$,

$$\Pr[\mathcal{A}(D) \in S] \leq e^\varepsilon \cdot \Pr[\mathcal{A}(D') \in S] + \delta.$$

If $\delta = 0$, we say that $\mathcal{A}$ is ε -DP.

In our privacy analysis, we use the following standard composition results, which capture how privacy loss accumulates when multiple DP algorithms are applied sequentially. A more general statement appears in [24].

THEOREM 3.2 (COMPOSITION [24, 25]). Let $\mathcal{A}_1, \dots, \mathcal{A}_k$ be ε -DP algorithms. Their k -fold adaptive composition $\mathcal{A}_{[k]}$, which outputs $y_i = \mathcal{A}_i(D, y_1, \dots, y_{i-1})$ for $i = 1, \dots, k$, satisfies:

- (i) $k\varepsilon$ -DP (Basic),
- (ii) $(\sqrt{2k \log(1/\delta)}\varepsilon + k\varepsilon(e^\varepsilon - 1), \delta)$ -DP for any $\delta > 0$ (Advanced).

In particular, to achieve (ε, δ) -DP for $\varepsilon \in (0, 1)$, it suffices that each $\mathcal{A}_i$ satisfies $\frac{\varepsilon}{2\sqrt{2k \log(1/\delta)}}\text{-DP}$.

DP algorithms are typically calibrated to the sensitivity of the underlying function with respect to single-record modifications of the dataset, defined formally as follows.

Definition 3.3 (Sensitivity [24]). The *sensitivity* of a function $q_D : \mathcal{R} \rightarrow \mathbb{R}$ that depends on a dataset D is defined as $\Delta_q = \sup_{r \in \mathcal{R}} \sup_{D \sim D'} |q_D(r) - q_{D'}(r)|$. In particular, the sensitivity of a set function $q_D : 2^V \rightarrow \mathbb{R}$ is $\sup_{S \subseteq V} \sup_{D \sim D'} |q_D(S) - q_{D'}(S)|$.

The exponential mechanism [41] is a primitive for privately selecting an approximately highest-scoring element from a candidate set according to a quality function that depends on the sensitive dataset.

Definition 3.4 (The Exponential Mechanism [41]). Given $D \in \mathcal{D}$, a finite set of candidates $\mathcal{R}$, a quality function $q_D : \mathcal{R} \rightarrow \mathbb{R}$, and a privacy parameter ε , the exponential mechanism $\text{EM}(D, q, \mathcal{R}, \varepsilon)$ outputs $r \in \mathcal{R}$ with probability proportional to $\exp\left(\frac{\varepsilon q_D(r)}{2\Delta_q}\right)$.

We assume that the sensitivity bound Δ_q is provided alongside q and do not consider it as a separate input.

THEOREM 3.5 ([6, 41]). *The exponential mechanism is ε -DP. Moreover,*

$$\mathbb{E}[\text{EM}(D, q, \mathcal{R}, \varepsilon)] \geq \max_{r \in \mathcal{R}} q_D(r) - \frac{2\Delta_q \log |\mathcal{R}|}{\varepsilon}.$$

The utility guarantee in expectation, which we use in this work, is due to [6]. The privacy guarantee is due to [41], as is the high-probability utility guarantee, which we omit here.

4 PROBLEM FORMULATION

The DP max-sum diversification (MSD) problem studied in this work extends the setting of Borodin et al. [8] to incorporate differential privacy requirements, and is formalized below.

Definition 4.1 (DP max-sum diversification). An instance of DP MSD is a tuple $\langle D, V, \mathcal{I}, f_D, d_D \rangle$ where:

- D is a sensitive dataset.
- V is a (public) dataset of items, called the *ground set*.
- $\mathcal{I} \subseteq 2^V$ is a collection of feasible subsets.
- $f_D : 2^V \rightarrow \mathbb{R}^+$ is a monotone submodular function that depends on the sensitive dataset D .
- $d_D : V \times V \rightarrow \mathbb{R}$ is a pseudometric (possibly depending on D), and we write $d_D(S) \triangleq \sum_{\{u,v\} \subseteq S} d_D(u,v)$.

The goal is to find $S \in \mathcal{I}$ that approximately maximizes

$$\phi_D(S) \triangleq f_D(S) + \lambda \cdot d_D(S)$$

while satisfying differential privacy with respect to D .

The parameter $\lambda \geq 0$ controls the trade-off between relevance and diversity. The subscript D is omitted when it is clear from context.

Remark 4.2. The relevance function f is submodular and the diversity term $d(S)$, being a sum of pairwise distances, is *supermodular*. Hence, the objective ϕ is not necessarily submodular.

Example 4.3. In the setting of Example 1.1, the sensitive dataset D consists of purchase records where each entry specifies the products purchased by a user. The ground set V contains Amazon products along with attributes such as product name, price, and category. The relevance function is defined as $f_D(S) = \frac{1}{m} |\bigcup_{v \in S} \mathcal{U}_D(v)|$, where $\mathcal{U}_D(v)$ is the set of users in D who purchased item v . Let $C(v)$ denote the set of categories of item $v \in V$, and the diversity is quantified by the Jaccard distance $d(v, v') = 1 - \frac{|C(v) \cap C(v')|}{|C(v) \cup C(v')|}$ between category sets of products v and v' .

Decomposable objective functions. A set function ϕ_D is called Δ -decomposable [33, 43] if it can be written as $\phi_D(S) = \sum_{x \in D} \phi_x(S)$, where each $\phi_x : 2^V \rightarrow [0, \Delta]$ depends only on the individual record x . A Δ -decomposable function has sensitivity Δ . Prior work has studied DP maximization of decomposable submodular functions [16, 17, 33], as they allow stronger privacy guarantees compared to the general case. Our algorithms for cardinality constraints similarly leverage this structure to provide privacy guarantees that nearly match known lower bounds for decomposable objectives.

Example 4.4. In the setting of Example 1.1, associate each user $x \in D$ with a monotone, submodular utility $f_x(S) = \min\{|S \cap \mathcal{P}(x)|, 1\}$, where $\mathcal{P}(x)$ is the set of products purchased by user x . This function

indicates whether user x is covered by S and depends solely on the data of that individual. For $\lambda \in [0, 1]$, define the per-user objective

$$\phi_x(S) = \frac{1-\lambda}{m} f_x(S) + \frac{2\lambda}{mk(k-1)} d(S)$$

where $m = |D|$. The normalization ensures that $\phi_x(S) \in [0, 1/m]$. The total objective is the average of these individual contributions:

$$\phi(S) = \sum_{x \in D} \phi_x(S) = \frac{1-\lambda}{m} \sum_{x \in D} f_x(S) + \frac{2\lambda}{k(k-1)} d(S).$$

Thus, ϕ is $1/m$ -decomposable. We now derive the sensitivity bound for ϕ . If $D = (D' \setminus \{x'\}) \cup \{x\}$, then for any $S \subseteq V$, $|\phi_D(S) - \phi_{D'}(S)| = |\phi_x(S) - \phi_{x'}(S)| \leq 1/m$, and thus ϕ has sensitivity bounded by $1/m$.

5 GREEDY ALGORITHM FOR CARDINALITY CONSTRAINTS

We start by presenting DP-Greedy, a DP adaptation of the greedy algorithm from [8], replacing each selection step with the exponential mechanism. Although it achieves high utility, as also shown empirically in Section 8, its complexity scales linearly in k , which can become prohibitive when k is large. Addressing this limitation constitutes a primary contribution of this work and is the focus of the subsequent section. DP-Greedy, stated as Algorithm 1, is *non-oblivious*: it selects elements according to the auxiliary function $\phi'_D(S) = \frac{1}{2}f_D(S) + \lambda d_D(S)$ rather than the objective ϕ_D .

Algorithm 1 DP-Greedy

Input: Dataset D , ground set V , monotone submodular f_D , pseudometric d_D , cardinality k , privacy parameter ε_0 , diversity parameter λ

Output: Size k subset of V

- 1: Define $\phi'_D(\cdot) = \frac{1}{2}f_D(\cdot) + \lambda d_D(\cdot)$
- 2: Set $S_0 \leftarrow \emptyset$
- 3: **for** $i = 1, \dots, k$ **do**
- 4: Let $N_i \leftarrow V \setminus S_{i-1}$
- 5: For all $u \in N_i$, define $q^i_D(u) = \phi'_D(u \mid S_{i-1})$
- 6: Compute $u_i \leftarrow \text{EM}(D, q^i, N_i, \varepsilon_0)$
- 7: $S_i \leftarrow S_{i-1} \cup \{u_i\}$
- 8: **end for**

THEOREM 5.1. Suppose ϕ_D has sensitivity Δ . Given parameter $\varepsilon_0 > 0$, DP-Greedy is $k\varepsilon_0$ -DP and (ε, δ) -DP for all $\delta > 0$ with $\varepsilon = \sqrt{2k \log(1/\delta)} \varepsilon_0 + k\varepsilon_0(e^{\varepsilon_0} - 1)$. Moreover, it outputs a feasible set S such that $\mathbb{E}[\phi_D(S)] \geq \frac{1}{2}\phi_D(\text{OPT}) - O(\frac{k\Delta \log n}{\varepsilon_0})$, and makes $O(nk)$ oracle calls.

In the special case where ϕ_D is decomposable (and hence so is ϕ'_D), DP-Greedy provides a stronger privacy guarantee compared to the general case, as stated next. The assumption that ϕ_D is 1-decomposable is without loss of generality, since we can apply our algorithms to the normalized function ϕ_D/Δ , and the guarantees for ϕ_D incur an additive error scaled by Δ .

THEOREM 5.2. Suppose ϕ'_D is 1-decomposable. Then given parameter $\varepsilon_0 \in (0, 1]$, DP-Greedy (Algorithm 1) is (ε, δ) -DP for every $\delta > 0$ and $\varepsilon = (e^{\varepsilon_0/2} - 1)(4 + \log(1/\delta))$.

Theorems 5.1 and 5.2 correspond to row (i) in Table 1. Informally, advanced composition implies that setting $\varepsilon_0 = O(\varepsilon/\sqrt{\log(1/\delta)k})$ ensures (ε, δ) -DP. However, Theorem 5.2 implies that setting $\varepsilon_0 = O(\varepsilon/\log(1/\delta))$ suffices to ensure (ε, δ) -DP, avoiding the dependence on k that arises from k compositions in the general case. Substituting this into the utility guarantee of Theorem 5.1 yields the stated additive error in Table 1. Theorem 5.2 is proved by the same steps as in the privacy proof for the CPP problem in [33]. Since this result follows as a special case of our more general analysis in Section 6, it is omitted. In the remainder of this section, we prove Theorem 5.1.

Let OPT be an optimal solution, i.e., a subset of size at most k that maximizes ϕ . Since ϕ is monotone, we may assume that $|\text{OPT}| = k$. Let S_i be the solution at the end of iteration i , and define $C_i = \text{OPT} \setminus S_{i-1}$. The approximation guarantee of the greedy algorithm in [8] relies on a lemma which relates the overall distance between the sets S_{i-1} and C_i to the value $d(\text{OPT})$. An adaptation of this lemma is stated next. The proof is deferred to Appendix B.

LEMMA 5.3 (ADAPTED FROM [8]). *For all $i \in [k] \triangleq \{1, \dots, k\}$, $d(C_i, S_{i-1}) \geq \frac{(i-1)|C_i|}{k(k-1)} d(\text{OPT})$.*

PROOF OF THEOREM 5.1. For the query complexity, observe that the algorithm performs k iterations, each requiring $O(n)$ oracle queries. The privacy guarantee follows directly from the composition theorems (Theorem 3.2), together with the ε_0 -DP property of the exponential mechanism.

Thus, we proceed with the utility guarantee. Let $f'(S) = f(S)/2$, so that $\phi'(S) = f'(S) + \lambda d(S)$. Consider any iteration $i \in [k]$, and let us condition on the set S_{i-1} selected after the first $i-1$ iterations. We have:

$$\begin{aligned} \sum_{u \in C_i} f'(u \mid S_{i-1}) &\geq f'(C_i \cup S_{i-1}) - f'(S_{i-1}) \\ &= f'(\text{OPT} \cup S_{i-1}) - f'(S_{i-1}) \geq f'(\text{OPT}) - f'(S_{i-1}), \end{aligned}$$

where the first inequality holds by the submodularity of f' , and the second by monotonicity. Moreover, by Lemma 5.3:

$$\sum_{u \in C_i} d(u \mid S_{i-1}) = d(C_i, S_{i-1}) \geq \frac{(i-1)|C_i|}{k(k-1)} d(\text{OPT}).$$

Letting $\alpha = 4\varepsilon_0^{-1} \Delta \log n$, we obtain:

$$\begin{aligned} \mathbb{E}[\phi'(u_i \mid S_{i-1})] &\geq \max_{u \in V \setminus S_{i-1}} \phi'(u \mid S_{i-1}) - \alpha \\ &\geq \frac{1}{|C_i|} \left(\sum_{u \in C_i} \phi'(u \mid S_{i-1}) \right) - \alpha \\ &\geq \frac{f'(\text{OPT}) - f'(S_{i-1})}{k} + \frac{\lambda(i-1)}{k(k-1)} d(\text{OPT}) - \alpha. \end{aligned}$$

The first inequality follows from the guarantees of the exponential mechanism, along with the fact that the quality functions q_D^i have sensitivity at most 2Δ . The second inequality holds since $C_i \subseteq V \setminus S_{i-1}$.

Removing the conditioning on S_{i-1} and taking the expectation over all its possible realizations, the last inequality yields:

$$\mathbb{E}[\phi'(u_i \mid S_{i-1})] \geq \frac{f'(\text{OPT}) - \mathbb{E}[f'(S_k)]}{k} + \frac{\lambda(i-1)}{k(k-1)} d(\text{OPT}) - \alpha,$$

where we have used the fact that $\mathbb{E}[f'(S_{i-1})] \leq \mathbb{E}[f'(S_k)]$ due to the monotonicity of f' . Summing over all $i \in [k]$, we get:

$$\mathbb{E}[\phi'(S_k)] \geq f'(\text{OPT}) - \mathbb{E}[f'(S_k)] + \frac{\lambda}{2} d(\text{OPT}) - k\alpha.$$

Plugging in $\phi' = f/2 + \lambda d$ and $f' = f/2$, we obtain:

$$\frac{1}{2} \mathbb{E}[f(S_k)] + \mathbb{E}[\lambda d(S_k)] \geq \frac{f(\text{OPT}) - \mathbb{E}[f(S_k)] + \lambda d(\text{OPT})}{2} - k\alpha.$$

Rearranging terms gives $\mathbb{E}[\phi(S_k)] \geq \frac{1}{2} \phi(\text{OPT}) - k\alpha$, which completes the proof. $\square$

6 FASTER GREEDY ALGORITHM VIA SUBSET SAMPLING

In this section, we present our main contribution for the cardinality-constraint setting. The Sample Greedy algorithm for monotone submodular maximization Buchbinder et al. [10], Mirzasoleiman et al. [42] achieves a $(1 - 1/e - \gamma)$ -approximation in expectation using only $O(n \log \gamma^{-1})$ oracle calls. Mitrovic et al. [43] proposed a variant for non-monotone submodular functions, guaranteeing a $(1 - 1/e)/e$ -approximation, and adapted it to satisfy differential privacy. However, these analyses rely on submodularity, which does not hold in our setting.

Nevertheless, we propose a Sample Greedy algorithm for DP MSD under cardinality constraints, stated as Algorithm 2. It achieves strong utility guarantees while using substantially fewer oracle calls than DP-Greedy. To the best of our knowledge, this is the fastest known algorithm for MSD with cardinality constraints, which is of interest even without privacy. Experimental results (Section 8) show that our method significantly improves running time over DP-Greedy while preserving utility. We further extend the analysis of Gupta et al. [33] to account for the subsampling step and obtain improved privacy guarantees for decomposable objectives.

Algorithm 2 DP-NOSG (DP Non-Oblivious Sample Greedy) / DP-OSG (DP Oblivious Sample Greedy)

Input: Dataset D , ground set V , submodular f_D , pseudometric d_D , cardinality k , privacy parameter ε_0 , utility parameter γ , diversity parameter λ . R1-D1

Output: Size k subset of V

- 1: In DP-NOSG: $\phi'_D(\cdot) = \frac{1}{2-\gamma} f_D(\cdot) + \lambda d_D(\cdot)$, $g(i) = k - i + 1$ for all $i \in [k]$ R1-D1
- 2: In DP-OSG: $\phi'_D(\cdot) = f_D(\cdot) + \lambda d_D(\cdot)$, $g(i) = \min\{k, n - i + 1\}$ for all $i \in [k]$ R1-D1
- 3: Set $S_0 \leftarrow \emptyset$
- 4: **for** $i = 1, \dots, k$ **do**
- 5: Set $N_i \leftarrow V \setminus S_{i-1}$
- 6: Sample a subset $V_i \subseteq N_i$ of size $\lceil |N_i| \cdot \min\left\{\frac{\log(1/\gamma)}{g(i)}, 1\right\} \rceil$ uniformly at random.
- 7: For all $u \in V_i$, define $q_D^i(u) = \phi'_D(u \mid S_{i-1})$
- 8: Compute $u_i \leftarrow \text{EM}(D, q_D^i, V_i, \varepsilon_0)$
- 9: $S_i \leftarrow S_{i-1} \cup \{u_i\}$
- 10: **end for**
- 11: **return** S_k

Algorithm 2 performs greedy selections with respect to a function ϕ' that can be either the objective ϕ or an auxiliary function,

and a utility function $g: [k] \rightarrow \mathbb{N}$, where $g(i)$ determines the size of the sampled candidate set in iteration i . We study two instantiations corresponding to different choices of g and ϕ' , yielding different trade-offs between running time and approximation guarantees.

6.1 Non-Oblivious Sample Greedy

The first instantiation we consider, referred to as DP-NOSG, selects the next element according to the auxiliary function $\phi'_D: 2^V \rightarrow \mathbb{R}$ given by $\phi'_D(S) = \frac{1}{2-\gamma} \cdot f_D(S) + \lambda d_D(S)$, and uses the utility function $g(i) = k - i + 1$ for all $i \in [k]$. We obtain the following result.

THEOREM 6.1. *Suppose ϕ_D has sensitivity Δ . Then, DP-NOSG (Algorithm 2) with parameters $\varepsilon_0 > 0$ and $\gamma \in (0, 1)$ is $k\varepsilon_0$ -DP and (ε, δ) -DP for all $\delta > 0$, where $\varepsilon = \sqrt{2k \log(1/\delta)} \varepsilon_0 + k\varepsilon_0(e^{\varepsilon_0} - 1)$. Moreover, it outputs a feasible set S such that $\mathbb{E}[\phi_D(S)] \geq (\frac{1}{2} - \gamma) \cdot \phi_D(\text{OPT}) - O\left(\frac{k\Delta \log n}{\varepsilon_0}\right)$ and makes $O(n \log k \log \gamma^{-1})$ oracle calls.*

By replacing the exponential mechanism with the true maximum, we get a faster, non-private algorithm for MSD, with the following guarantees.

COROLLARY 6.2. *Instantiated with MAX instead of EM, DP-NOSG outputs a feasible set S such that $\mathbb{E}[\phi(S)] \geq (\frac{1}{2} - \gamma)\phi(\text{OPT})$ and makes $O(n \log k \log \gamma^{-1})$ oracle calls.*

The next theorem states the improved privacy guarantees for decomposable objective functions. Note that under our definition of ϕ' , if ϕ is 1-decomposable, then so is ϕ' .

THEOREM 6.3. *Suppose $\phi_D: 2^V \rightarrow \mathbb{R}$ is 1-decomposable. Then given parameter $\varepsilon_0 \in (0, 1]$, Algorithm 2 is (ε, δ) -DP for every $\delta > 0$ and $\varepsilon = (e^{\varepsilon_0/2} - 1)(4 + \log(1/\delta))$.*

Theorems 6.1 and 6.3 correspond to row (ii) in Table 1. As in the previous section, Theorem 5.2 implies that for decomposable objectives, setting $\varepsilon_0 = O(\varepsilon/\log(1/\delta))$ suffices to ensure (ε, δ) -DP. Substituting this into the utility guarantee of Theorem 6.1 yields the stated additive error in Table 1.

Before proceeding to the proof, we introduce additional notation. Let OPT be an optimal solution and S_i the set obtained at the end of iteration i . Recall that $N_i = V \setminus S_{i-1}$ contains all unselected elements after iteration $i-1$, and $C_i = \text{OPT} \setminus S_{i-1}$ denotes the subset of these elements belonging to OPT. Let $M_i \subseteq N_i$ be a set of size $g(i)$ maximizing $\sum_{v \in M_i} \phi'(v \mid S_{i-1})$. That is, M_i consists of the $g(i)$ elements with the largest marginal contributions during iteration i relative to the auxiliary function ϕ' . We prove the following.

LEMMA 6.4. *For every iteration $i \in [k]$, conditioned on having selected a set S_{i-1} after the first $i-1$ iterations, the following holds. $\mathbb{E}[\phi'(u_i \mid S_{i-1})] \geq \frac{1-\gamma}{|M_i|} \sum_{v \in M_i} \phi'(v \mid S_{i-1}) - \frac{4\Delta \log n}{\varepsilon_0}$*

PROOF. Let $\alpha = 4\varepsilon_0^{-1} \Delta \log n$ and $i \in [k]$. Condition on a realization of S_{i-1} and V_i . By Theorem 3.5 and the 2Δ sensitivity bound for q_D^i , we have

$$\begin{aligned} \mathbb{E}[\phi'(u_i \mid S_{i-1})] &\geq \max_{v \in V_i} \phi'(v \mid S_{i-1}) - \alpha \geq \max_{v \in V_i \cap M_i} \phi'(v \mid S_{i-1}) - \alpha \\ &\geq \sum_{v \in V_i \cap M_i} \frac{\phi'(v \mid S_{i-1})}{|V_i \cap M_i|} - \alpha. \end{aligned}$$

Note that second inequality holds even if $V_i \cap M_i = \emptyset$ because ϕ' is monotone and marginal gains are nonnegative. Removing the conditioning on V_i and taking the expectation over all its realizations gives

$$\begin{aligned} \mathbb{E}[\phi'(u_i \mid S_{i-1})] &\geq \mathbb{E}\left[\sum_{v \in V_i \cap M_i} \frac{\phi'(v \mid S_{i-1})}{|V_i \cap M_i|}\right] - \alpha \\ &\geq \frac{1-\gamma}{|M_i|} \sum_{v \in M_i} \phi'(v \mid S_{i-1}) - \alpha. \end{aligned} \quad (1)$$

To prove the last inequality, let G be the event $V_i \cap M_i \neq \emptyset$. If $\log(1/\gamma) \geq g(i)$, then $\Pr[G] = 1$ holds. Otherwise, the sampling in Line 6 ensures

$$\Pr[G] = \Pr[V_i \cap M_i \neq \emptyset] \geq 1 - \left(1 - \frac{g(i)}{|N_i|}\right)^{\frac{|N_i| \log(1/\gamma)}{g(i)}} \geq 1 - \gamma.$$

Let p_v be the probability that $v \in M_i$ appears in V_i conditioned on G . By the law of total expectation:

$$\begin{aligned} \mathbb{E}\left[\sum_{v \in V_i \cap M_i} \frac{\phi'(v \mid S_{i-1})}{|V_i \cap M_i|}\right] &= \Pr[G] \cdot \sum_{v \in M_i} p_v \phi'(v \mid S_{i-1}) \\ &\geq \frac{1-\gamma}{|M_i|} \sum_{v \in M_i} \phi'(v \mid S_{i-1}) \end{aligned}$$

where the first equality uses the fact that the inner sum is zero conditioned on $\bar{G}$. Because V_i is chosen uniformly, p_v is identical for all $v \in M_i$, thus $p_v = 1/|M_i|$. This yields (1). $\square$

PROOF OF THEOREM 6.1. The complexity bound holds due to the fact that DP-NOSG evaluates $O(\frac{n}{k-i} \log \gamma^{-1})$ candidates in iteration i , the number of oracle calls across k iterations is $O(n \log k \log \gamma^{-1})$ using the standard Harmonic sum approximation. The full argument is deferred to Appendix C. Privacy follows from the composition theorems (Theorem 3.2) and the ε_0 -DP guarantee of the exponential mechanism. We thus focus on the utility guarantee.

Let $f' = \frac{1}{2-\gamma} f$ and $\phi' = f' + \lambda d$. Define $\alpha = 4\varepsilon_0^{-1} \Delta \log n$. Condition on having selected S_{i-1} prior to iteration $i \in [k]$. By Lemma 6.4, we have

$$\begin{aligned} \mathbb{E}[\phi'(u_i \mid S_{i-1})] &\geq \frac{1-\gamma}{|M_i|} \sum_{v \in M_i} \phi'(v \mid S_{i-1}) - \alpha \\ &\geq \frac{1-\gamma}{|C_i|} \sum_{v \in C_i} \phi'(v \mid S_{i-1}) - \alpha \\ &\geq (1-\gamma) \left(\frac{f'(\text{OPT}) - f'(S_{i-1})}{k} + \frac{\lambda(i-1)}{k(k-1)} d(\text{OPT}) \right) - \alpha. \end{aligned}$$

The second inequality holds because $|M_i| = k - i + 1 \leq |C_i|$, and M_i contains the elements with the largest marginal gains. The third follows from submodularity of f' and Lemma 5.3 for d . Taking an expectation over S_{i-1} gives

$$\mathbb{E}[\phi'(u_i \mid S_{i-1})] \geq (1-\gamma) \left(\frac{f'(\text{OPT}) - \mathbb{E}[f'(S_k)]}{k} + \frac{\lambda(i-1)}{k(k-1)} d(\text{OPT}) \right) - \alpha.$$

where we use the monotonicity of f' and the fact that $S_{i-1} \subseteq S_k$ to bound $\mathbb{E}[f'(S_k)] \geq \mathbb{E}[f'(S_{i-1})]$. Summing over $i \in [k]$ yields

$$\mathbb{E}[\phi'(S_k)] \geq (1-\gamma) (f'(\text{OPT}) - \mathbb{E}[f'(S_k)] + \frac{\lambda}{2} d(\text{OPT})) - k\alpha.$$

Rearranging terms and substituting f' gives:

$$\mathbb{E}[\phi(S_k)] \geq (1-\gamma) \left(\frac{f(\text{OPT})}{2-\gamma} + \frac{\lambda d(\text{OPT})}{2} \right) - k\alpha \geq (\frac{1}{2} - \gamma) \phi(\text{OPT}) - k\alpha. \quad \square$$

Next, we address the stronger guarantee of Theorem 6.3, which applies when the function ϕ is decomposable. We build on the technique of [33], which bounds the privacy loss of the greedy algorithm for submodular maximization by the sum of expected marginal gains of a single individual's function f_x across all rounds, where the expectation is taken over a distribution induced by the functions of the other individuals. This quantity is bounded with high probability by the sum of realized marginal gains $\sum_{i=1}^k [f_x(S_i) - f_x(S_{i-1})] = f_x(S_k) - f_x(S_0) \leq \Delta$, which telescopes to a constant independent of k . We extend this technique to Algorithm 2 by providing a unified analysis for the subsampling steps and the EM. Specifically, we prove that for any deterministic and adaptive choice of V_i , the privacy loss can be bounded by a sum of expected marginal gains, which is bounded by a telescoping sum with high probability, analogous to [33]. This extension enables our utility bound to outperform a naive composition-based analysis of repeated EM, and approach existing lower bounds. We provide a proof sketch, with the full proof given in Appendix C.

PROOF OF THEOREM 6.3. Let D and D' be two datasets such that $(D \setminus D') \cup (D' \setminus D) = \{x\}$. Suppose that instead of a set, Algorithm 2 outputs the sequence of selected elements in their order of selection. Let $U = (u_1, \dots, u_k)$ be any such sequence, and $U_i = \{u_1, \dots, u_i\}$ denote the prefix set consisting of the first i elements.

For any element $u \in V$, let $\beta_D^i(u) = \sum_{x \in D} \phi'_x(u \mid U_{i-1})$. By the definition of the exponential mechanism, the probability of selecting element u at iteration i , given dataset D and a sampled candidate set V_i , is:

$$\Pr[\text{EM}(q_i, V_i, D, \varepsilon_0) = u] = \frac{\exp(\frac{\varepsilon_0}{2} \beta_D^i(u))}{\sum_{u' \in V_i} \exp(\frac{\varepsilon_0}{2} \beta_D^i(u'))}.$$

where we have used that since ϕ'_D is 1-decomposable, it has a sensitivity $1/|D|$. The expression for D' follows analogously. Let $\mathcal{G}$ denote Algorithm 2. By the chain rule, the probability of $\mathcal{G}$ outputting sequence U is:

$$\Pr[\mathcal{G}(D) = (u_1, \dots, u_k)] = \prod_{i=1}^k \Pr[\mathcal{G}(D)_i = u_i \mid S_{i-1} = U_{i-1}].$$

Let $\mathcal{V}_i$ denote the collection of all subsets of N_i of the size specified in Line 6. Since V_i is sampled uniformly at random from $\mathcal{V}_i$, for every $T \in \mathcal{V}_i$, we have $\Pr[V_i = T] = p_i$, where $p_i = 1/|\mathcal{V}_i|$ is independent of the dataset D . By the law of total probability, the i -th factor in the product above is:

$$\Pr[\mathcal{G}(D)_i = u_i \mid S_{i-1} = U_{i-1}] = \sum_{T \in \mathcal{V}_i: u_i \in T} \Pr[\text{EM}(q_i, T, D, \varepsilon_0) = u_i] \cdot p_i.$$

Note that the selection probability is zero if $u_i \notin V_i$. Using the fact that $\frac{\sum_j a_j}{\sum_j b_j} \leq \max \frac{a_j}{b_j}$ for $a_j \geq 0$ and $b_j > 0$, we have:

$$\begin{aligned} \frac{\Pr[\mathcal{G}(D)_i = u_i \mid S_{i-1} = U_{i-1}]}{\Pr[\mathcal{G}(D')_i = u_i \mid S_{i-1} = U_{i-1}]} &= \frac{\sum_{T \in \mathcal{V}_i: u_i \in T} \Pr[\text{EM}(q_i, T, D, \varepsilon_0) = u_i]}{\sum_{T \in \mathcal{V}_i: u_i \in T} \Pr[\text{EM}(q_i, T, D', \varepsilon_0) = u_i]} \\ &\leq \max_{T \in \mathcal{V}_i: u_i \in T} \left[\frac{\Pr[\text{EM}(q_i, T, D, \varepsilon_0) = u_i]}{\Pr[\text{EM}(q_i, T, D', \varepsilon_0) = u_i]} \right]. \end{aligned}$$

Let T_i be the candidate set attaining the maximum above. Taking the product over all iterations k :

$$\frac{\Pr[\mathcal{G}(D) = U]}{\Pr[\mathcal{G}(D') = U]} \leq \left(\prod_{i=1}^k \frac{\exp(\frac{\varepsilon_0}{2} \beta_D^i(u_i))}{\exp(\frac{\varepsilon_0}{2} \beta_{D'}^i(u_i))} \right) \left(\prod_{i=1}^k \frac{\sum_{u \in T_i} \exp(\frac{\varepsilon_0}{2} \beta_{D'}^i(u))}{\sum_{u \in T_i} \exp(\frac{\varepsilon_0}{2} \beta_D^i(u))} \right). \quad (2)$$

We consider two cases.

Case 1: $D = D' \cup \{x\}$. The first factor in (2) is bounded by

$$\exp\left(\frac{\varepsilon_0}{2} \sum_{i=1}^k \phi'_x(u_i \mid U_{i-1})\right) \leq \exp\left(\frac{\varepsilon_0}{2}\right)$$

due to 1-decomposability. The second factor is at most 1 since $\beta_D^i(u) \geq \beta_{D'}^i(u)$.

Case 2: $D' = D \cup \{x\}$. The first factor is at most 1, while the second factor becomes:

$$\prod_{i=1}^k \frac{\sum_{u \in T_i} \exp(\frac{\varepsilon_0}{2} \phi'_x(u \mid U_{i-1})) \exp(\frac{\varepsilon_0}{2} \beta_D^i(u))}{\sum_{u \in T_i} \exp(\frac{\varepsilon_0}{2} \beta_D^i(u))} \quad (3)$$

$$= \prod_{i=1}^k \mathbb{E}_{u \sim P_i} \left[\exp\left(\frac{\varepsilon_0}{2} \phi'_x(u \mid U_{i-1})\right) \right], \quad (4)$$

where P_i is a distribution supported on T_i with weights proportional to $\exp(\frac{\varepsilon_0}{2} \beta_D^i(u))$. To bound this product of expectations, we apply the concentration bound from [16, 33]. By Lemma C.3, with probability at least $1 - \delta$:

$$\frac{\Pr[\mathcal{G}(D) = U]}{\Pr[\mathcal{G}(D') = U]} \leq \exp\left((e^{\varepsilon_0/2} - 1)(3 + \log(1/\delta))\right).$$

Combining both cases, for any neighboring D, D' (under the change-one-element definition), with probability at least $1 - \delta$:

$$\begin{aligned} \frac{\Pr[\mathcal{G}(D) = U]}{\Pr[\mathcal{G}(D') = U]} &\leq \exp\left(\frac{\varepsilon_0}{2} + (e^{\varepsilon_0/2} - 1)(3 + \log(1/\delta))\right) \\ &\leq \exp\left((e^{\varepsilon_0/2} - 1)(4 + \log(1/\delta))\right). \end{aligned}$$

Thus, Algorithm 2 is (ε, δ) -DP with $\varepsilon = (e^{\varepsilon_0/2} - 1)(4 + \log(1/\delta))$. $\square$

So far, we have analyzed DP-NOSG, an instantiation of Algorithm 2 that makes $O_\gamma(n \log k)$ oracle calls. In the following section, we introduce a linear-time algorithm where the number of oracle calls is independent of k .

6.2 Oblivious Sample Greedy

We present DP-OSG, an efficient algorithm for DP MSD that makes only $O_\gamma(n)$ oracle calls, albeit with a weaker approximation factor. Our experiments in Section 8 demonstrate that DP-OSG achieves high utility under tight privacy constraints. Furthermore, it is significantly more scalable than DP-Greedy, making it better suited for interactive settings. DP-OSG uses the utility function $g(i) = \min\{k, n - i + 1\}$ for all $i \in [k]$. A key component in the analysis of DP-NOSG is Lemma 5.3, which lower bounds the total distance between the current solution and the remaining unselected elements of OPT as a function of $d(\text{OPT})$. While this yields tight guarantees for DP-Greedy and DP-NOSG, applying the same technique with the current choice of g leads only to a $1/6 - \gamma$ approximation. We therefore take a different approach and instantiate Algorithm 2 with $\phi' = \phi$, i.e., in an oblivious manner. Instead of relying on Lemma 5.3, we bound the total distance to the remaining elements of OPT using

the marginal gain $d(\text{OPT} \cup S_i) - d(S_i)$ (Lemma 6.7). This yields a new progress inequality for the oblivious variant, leading to a tighter approximation factor.

One can verify that the privacy guarantees stated in Theorems 6.1 and 6.3 hold for Algorithm 2 with the current choice of g and ϕ' .

THEOREM 6.5. *Suppose ϕ has sensitivity Δ . Then, DP-OSG (Algorithm 2) outputs a feasible set S such that*

$$\mathbb{E}[\phi(S)] \geq (1 - (\frac{2}{e})^{(1-\gamma)(1-1/k)}) \cdot \phi(\text{OPT}) - O(\frac{k\Delta \log n}{\epsilon_0})$$

and makes $O(n \log \gamma^{-1})$ oracle calls.

Lemma C.4 shows that the approximation factor is greater than $1 - 2/e - \gamma - 1/k$, and we use this linearized form for simplicity. Theorem 6.5, together with the privacy guarantees in Theorems 6.1 and 6.3 correspond to row (iii) in Table 1. By replacing the exponential mechanism with the true maximum, we get the first linear-time, non-private algorithm with a constant factor approximation.

COROLLARY 6.6. *Instantiated with MAX instead of EM, DP-OSG outputs a feasible set S such that $\mathbb{E}[\phi(S)] \geq (1 - (\frac{2}{e})^{(1-\gamma)(1-1/k)}) \cdot \phi(\text{OPT})$, and makes $O(n \log \gamma^{-1})$ oracle queries.*

The proof of Theorem 6.5 relies on the following structural progress lemma, whose proof is deferred to the full version [60].

LEMMA 6.7. *For every iteration $i \geq 2$,*

$$d(\text{OPT} \cup S_{i-1}) - d(S_{i-1}) \leq \left(1 + \frac{k-1}{i-1}\right) \cdot d(C_i, S_{i-1}).$$

PROOF OF THEOREM 6.5. The proof of the query complexity is provided in Lemma C.1. We thus focus on the utility guarantee. Let $\alpha = 4\Delta \log(n)/\epsilon_0$. Consider any iteration $2 \leq i \leq k$ and condition on the set S_{i-1} selected after iteration $i-1$. By the submodularity and monotonicity of f , we have:

$$\sum_{u \in C_i} f(u \mid S_{i-1}) \geq f(\text{OPT} \cup S_{i-1}) - f(S_{i-1}) \geq f(\text{OPT}) - f(S_{i-1}).$$

Applying Lemma 6.7, we find that:

$$\sum_{v \in C_i} d(v \mid S_{i-1}) = d(C_i, S_{i-1}) \geq \frac{d(\text{OPT} \cup S_{i-1}) - d(S_{i-1})}{1 + \frac{k-1}{i-1}}.$$

Let $M_i \subseteq N_i$ be a set of size $g(i) = \min\{k, |N_i|\}$ that maximizes $\sum_{v \in M_i} \phi(v \mid S_{i-1})$. Note that M_i comprises the k elements with the largest marginal contributions in iteration i , or is equal to N_i if $|N_i| \leq k$. We have:

$$\begin{aligned} \mathbb{E}[\phi(u_i \mid S_{i-1})] &\geq \frac{1-\gamma}{g(i)} \sum_{v \in M_i} \phi(v \mid S_{i-1}) - \alpha \geq \frac{1-\gamma}{k} \sum_{v \in C_i} \phi(v \mid S_{i-1}) - \alpha \\ &\geq (1-\gamma) \left(\frac{\phi(\text{OPT}) - \phi(S_{i-1})}{k(1 + \frac{k-1}{i-1})} \right) - \alpha. \end{aligned}$$

The first inequality follows from the guarantees of the exponential mechanism and the sampling procedure, analogous to the proof of Lemma 6.4. For the second inequality, recall that $C_i \subseteq N_i$, and since $C_i \subseteq \text{OPT}$, it holds that $|C_i| \leq k$. Thus, the inequality follows from the definition of M_i , together with the fact that $|C_i| \leq g(i) \leq k$.

Removing the conditioning on S_{i-1} and taking the expectation over all possible realizations yields:

$$\mathbb{E}[\phi(u_i)] \geq (1-\gamma) \left(\frac{\phi(\text{OPT}) - \mathbb{E}[\phi(S_{i-1})]}{k(1 + \frac{k-1}{i-1})} \right) - \alpha.$$

Rearranging terms, we obtain:

$$\phi(\text{OPT}) - \mathbb{E}[\phi(S_i)] \leq \left(1 - \frac{1-\gamma}{k(1 + \frac{k-1}{i-1})}\right) (\phi(\text{OPT}) - \mathbb{E}[\phi(S_{i-1})]) + \alpha.$$

By recursively applying this inequality, we get:

$$\begin{aligned} \phi(\text{OPT}) - \mathbb{E}[\phi(S_k)] &\leq \prod_{i=2}^k \left(1 - \frac{1-\gamma}{k(1 + \frac{k-1}{i-1})}\right) \phi(\text{OPT}) + k\alpha \\ &\leq \left(\frac{2}{e}\right)^{(1-\gamma)(1-1/k)} \phi(\text{OPT}) + k\alpha, \end{aligned}$$

where the last inequality follows from Lemma C.4. The theorem follows by rearranging the final expression. $\square$

7 LOCAL SEARCH ALGORITHM FOR MATROID CONSTRAINTS

In this section, we present our algorithm for matroid constraints. Borodin et al. [8] proposed a non-private local search achieving a $(1/2 - \gamma)$ -approximation in $O(n^2 + \gamma^{-1}nk^2 \log k)$ oracle calls³ starting from an initial base containing the highest-scoring feasible pair of elements. However, privatizing this approach is non-trivial: additive DP noise prevents both verifying local optimality and guaranteed selection of strictly improving swaps.

Our algorithm, DP-SLS (Algorithm 3), executes a fixed number of local search iterations, using the exponential mechanism to privately select an approximately best available swap from a subsampled candidate set. To avoid forced suboptimal swaps when no improving swap exists, we introduce a dummy swap that allows the current solution to remain unchanged. Finally, we apply the exponential mechanism to privately select and output the highest-scoring solution observed throughout the execution.

Our analysis exploits the structure of the underlying metric space to establish sufficient expected progress and achieve a near-optimal approximation ratio, while avoiding higher-order error terms introduced by continuous relaxations [49]. We show that whenever the expected objective value in a given iteration is low, the subsequent iteration increases it significantly. In particular, the first iteration already guarantees a sufficiently large expected objective value, enabling a multiplicative progress argument that yields a $(1/2 - \gamma)$ -approximation within $O(\gamma^{-1}k \log k)$ iterations.

By basing our approach on local search, we depart from predominantly greedy methods used in DP submodular maximization. This approach yields a combinatorial proof with a tighter additive error term, while simultaneously eliminating the $O(n^2)$ factor present in the complexity of [8]. We further reduce complexity by subsampling candidate swaps in each iteration, ultimately requiring

³The algorithm in [8] attains a $1/2$ -approximation but is not polynomial-time. The authors' proposal to perform only swaps yielding at least a γ -improvement at each iteration, rather than any improvement, leads to the stated guarantees.

$O(\gamma^{-1}nk \log k)$ oracle calls. This provides a substantial improvement over prior work while maintaining the same near-optimal expected approximation factor.

THEOREM 7.1. *Suppose $\phi_D : 2^V \rightarrow \mathbb{R}$ has sensitivity Δ , and let $\mathcal{M}$ be a matroid of rank k . For $T' = O(\gamma^{-1}k \log k)$, DP-SLS (Algorithm 3) with parameters $\varepsilon_0 > 0$ and $\gamma \in (0, 1)$ is $T'\varepsilon_0$ -DP, and (ε, δ) -DP for every $\delta > 0$, with $\varepsilon = \sqrt{2T' \log(1/\delta)}\varepsilon_0 + T'\varepsilon_0(e^{\varepsilon_0} - 1)$. Moreover, it outputs a feasible set S such that*

$$\mathbb{E}[\phi_D(S)] \geq \left(\frac{1}{2} - \gamma\right) \cdot \phi_D(\text{OPT}) - O\left(\frac{\Delta}{\varepsilon_0} \left(k \log n + \log \frac{1}{\gamma}\right)\right),$$

and makes $O(\gamma^{-1}nk \log k)$ oracle calls.

R2-W2

Theorem 7.1 corresponds to row (iv) in Table 1. The stated additive error in Table 1 is obtained using advanced composition (Theorem 3.2), which intuitively implies that setting $\varepsilon_0 = O(\varepsilon/\sqrt{\gamma^{-1}k \log k \log \delta^{-1}})$ ensures (ε, δ) -DP. Note that we now have $O(\gamma^{-1}k \log k)$ compositions. By replacing the exponential mechanism with the true maximum, we get a faster, non-private algorithm for the MSD problem with matroid constraints, for which the expected approximation guarantee is the same as that of the deterministic local search algorithm of [8].

COROLLARY 7.2. *Instantiated with MAX instead of EM, DP-SLS outputs a feasible set S such that $\mathbb{E}[\phi(S)] \geq (\frac{1}{2} - \gamma) \cdot \phi(\text{OPT})$ and makes $O(\gamma^{-1}nk \log k)$ oracle calls.*

Remark 7.3. In this section, we consider the broader class of Δ -sensitive functions. The technique of Gupta et al. [33] for decomposable functions relies crucially on the monotonic growth of the greedy solution sequence. Extending this approach to local-search algorithms, where elements are both added and removed, appears nontrivial and remains an interesting direction for future work.

Algorithm 3 DP Sample Local Search (DP-SLS)

Input: Dataset D , ground set V , submodular f_D , pseudometric d_D , matroid $\mathcal{M}$ of rank k , privacy parameter ε_0 , utility parameter $\gamma \in (0, 1]$, **diversity parameter λ**

R1-D1

- 1: **Let** $\phi_D(\cdot) = f_D(\cdot) + \lambda d_D(\cdot)$
 - 2: Let S_0 be an arbitrary base of $\mathcal{M}$.
 - 3: Set $T \leftarrow \lceil \frac{2k \log(8k)}{\gamma(1-1/e)} \rceil + 1$
 - 4: **for** $i = 1, \dots, T$ **do**
 - 5: Sample a subset $V_i \subseteq V$ of size $\lceil \frac{n}{k} \rceil$ uniformly at random
 - 6: Let $W_i \leftarrow \{(u, v) \in S_{i-1} \times (V_i \setminus S_{i-1}) \mid S_{i-1} - u + v \in I\} \cup \{(w, w)\}$ for an arbitrary element $w \in S_{i-1}$.
 - 7: For all $(u, v) \in W_i$, define $q_D^i(u, v) = \phi_D(S_{i-1} - u + v)$.
 - 8: Compute $(u_i, v_i) \leftarrow \text{EM}(D, q_D^i, W_i, \varepsilon_0)$
 - 9: Let $S_i \leftarrow S_{i-1} - u_i + v_i$
 - 10: **end for**
 - 11: For all $i = 1, \dots, T$, define $s_D(i) = \phi_D(S_i)$.
 - 12: Compute $i^* \leftarrow \text{EM}(D, s_D, [T], \varepsilon_0)$
 - 13: **return** S_{i^*}
-

For the remainder of this section we prove Theorem 7.1. For simplicity, we assume throughout that the $\mathcal{M}$ has rank $k \geq 3$, and address the case $k = 2$ in Appendix E.

Given a set S and elements u, v , we use $S + u$, $S - u$ and $S - u + v$ as shorthands for $S \cup \{u\}$, $S \setminus \{u\}$ and $(S \setminus \{u\}) \cup \{v\}$ respectively. Let OPT be an optimal solution. Since ϕ is monotone, we may assume, without loss of generality, that OPT is a base of $\mathcal{M}$. For $i \in [T]$, let S_{i-1} denote the base obtained after iteration $i - 1$.

LEMMA 7.4. *Suppose $\mathcal{M}$ has rank $k > 2$. There exists a bijection $h : S_{i-1} \rightarrow \text{OPT}$ such that $h(u) = u$ for every $u \in S_{i-1} \cap \text{OPT}$, and $\text{OPT} - h(u) + u \in \mathcal{I}$ for every $u \in S_{i-1}$. Moreover,*

$$\sum_{u \in S_{i-1}} \phi(S_{i-1} - u + h(u)) \geq \phi(\text{OPT}) + (k - 2)\phi(S_{i-1}).$$

Using this lemma, we establish the following result, characterizing the algorithm's progress.

LEMMA 7.5. *For every iteration $i \in [T]$, we have*

$$\mathbb{E}[\phi(S_i) - \phi(S_{i-1})] \geq \frac{1-1/e}{k} (\phi(\text{OPT}) - 2\mathbb{E}[\phi(S_{i-1})]) - \frac{4\Delta \log n}{\varepsilon_0}$$

Intuitively, Lemma 7.4 implies that after obtaining a base S_{i-1} , a uniformly random swap from the set $M_i = \{(u, h(u)) : u \in S_{i-1}\}$ yields an expected gain of $\frac{1}{k}(\phi(\text{OPT}) - 2\phi(S_{i-1}))$. To prove Lemma 7.5, we show that the set of swaps considered at iteration i intersects M_i with probability at least $1 - 1/e$. Conditioned on this event, the swap selected by the exponential mechanism achieves, up to an additive error, the maximal gain available, which in turn is at least as large as the average gain of a swap in M_i . Combining these observations establishes the lemma. The full argument, which also handles potential negative gains, is presented in Appendix D.

PROOF OF THEOREM 7.1. The privacy guarantee follows directly from the ε_0 -DP of the exponential mechanism and the composition theorems (Theorem 3.2), noting that we now have $T' \triangleq T + 1 = O(\gamma^{-1}k \log k)$ compositions. For the query complexity, Algorithm 3 performs T iterations, and each iteration requires only $O(|S_{i-1}| \cdot |V_i|) = O(n)$ queries due to the sub-sampling step. Overall, the algorithm performs $O(\gamma^{-1}nk \log k)$ queries. We proceed to prove the utility guarantee. Let $\alpha = 4\Delta \log(n)/\varepsilon_0$. We may assume that $\frac{1-1/e}{k} \cdot \phi(\text{OPT}) - \alpha \geq \frac{\phi(\text{OPT})}{8k}$, since otherwise, it can be verified that $\phi(\text{OPT}) < 2k\alpha$, and the guarantees of Theorem 7.1 hold trivially. Let S_0 be the first base chosen by the algorithm. We have

$$\begin{aligned} \mathbb{E}[\phi(S_1)] &\geq \phi(S_0) + \frac{1-1/e}{k} (\phi(\text{OPT}) - 2\phi(S_0)) - \alpha \\ &\geq \frac{1-1/e}{k} \cdot \phi(\text{OPT}) - \alpha \geq \frac{\phi(\text{OPT})}{8k} \end{aligned}$$

where the first inequality holds by Lemma 7.5, and the second since $k \geq 2$. Now, fix an iteration $2 \leq i \leq T$, and let us condition on having selected some base S_{i-1} after the first $i - 1$ iterations. If $\mathbb{E}[\phi(S_{i-1})] < \phi(\text{OPT})/(2 + \gamma) - k\alpha$, then, by Lemma 7.5 and since $\alpha \geq 0$, we obtain

$$\mathbb{E}[\phi(S_i)] \geq \left(1 + \frac{\gamma(1-1/e)}{k}\right) \mathbb{E}[\phi(S_{i-1})] \quad (5)$$

We claim that given our choice of T , there must exist some $i \in [T]$ for which $\mathbb{E}[\phi(S_i)] \geq \phi(\text{OPT})/(2 + \gamma) - k\alpha$. Assume for contradiction that this is not the case. Recursively applying the bound (5)

yields

$$\begin{aligned}\mathbb{E}[\phi(S_T)] &\geq \left(1 + \frac{\gamma(1-1/e)}{k}\right)^{\frac{2k \log(8k)}{\gamma(1-1/e)}} \mathbb{E}[\phi(S_1)] \\ &\geq \frac{1}{8k} \left(1 + \frac{\gamma(1-1/e)}{k}\right)^{\frac{2k \log(8k)}{\gamma(1-1/e)}} \phi(\text{OPT}) \\ &> \phi(\text{OPT}).\end{aligned}$$

The last inequality from the bound $(1 + 1/x)^x \geq \exp(1 - 1/(2x))$ which holds for all $x > 0$. In particular, there exists some base S with $\phi(S) > \phi(\text{OPT})$, in contradiction. Thus, by the law of total expectation, we get

$$\begin{aligned}\mathbb{E}[\phi(S_i^*)] &= \mathbb{E}[\mathbb{E}[\phi(S_i^*) \mid S_1, \dots, S_T] \geq \mathbb{E}\left[\max_{i \in [T]} \phi(S_i)\right] - \frac{2\Delta \log T}{\varepsilon_0} \\ &\geq \max_{i \in [T]} \mathbb{E}[\phi(S_i)] - \frac{2\Delta \log T}{\varepsilon_0} \geq \frac{\phi(\text{OPT})}{2+\gamma} - k\alpha - \frac{2\Delta \log T}{\varepsilon_0} \\ &= \frac{\phi(\text{OPT})}{2+\gamma} - \frac{2\Delta}{\varepsilon_0} \cdot \left(2k \log n + \log\left(\frac{2k \log k}{\gamma(1-1/e)}\right)\right)\end{aligned}$$

The second inequality holds by the fact that $\mathbb{E}[\max_i X_i] \geq \max_i \mathbb{E}[X_i]$ for any jointly distributed random variables $X_1, \dots, X_n$. As $1/(2 + \gamma) \geq 1/2 - \gamma$, the theorem follows. $\square$

8 EXPERIMENTS

In this section, we evaluate the utility and efficiency of our proposed algorithms across two data summarization applications: Uber pickup location summarization and Amazon product summarization. Based on these applications, we characterize the trade-offs between privacy, utility, and complexity.

Summary of Findings. For cardinality constraints, the proposed DP algorithms achieve competitive utility compared to the non-private baseline while drastically improving efficiency. DP-Greedy and DP-NOSG stay within a marginal 1% of the non-private baseline even at a strict $\varepsilon = 0.1$ budget. By shifting the query complexity to logarithmic or zero dependence on k , DP-NOSG and DP-OSG achieve speedups of 14 $\times$ and 50 $\times$ at $k = 100$. For matroid constraints, DP-SLS maintains competitive utility, staying within 1% of the non-private baseline at $\varepsilon = 0.1$. While DP-SLS reduces the total number of oracle calls, its fixed iteration count leads to an execution time overhead of up to 2.2 $\times$ compared to the non-private baseline; however, despite scaling more slowly with k , it exhibits the improved scaling with n predicted by our analysis.

8.1 Experimental Settings

We next present our settings for the experiments. All algorithms were implemented⁴ in Python 3.9.19 using the Pandas and NumPy libraries. All experiments were run on an Intel Xeon CPU-based server with 24 cores and 96 GB of RAM.

Default parameters. Unless mentioned otherwise, the following parameters are used. We set $\varepsilon = 0.1$ and $\delta = |D|^{-1.5}$ where D is the sensitive dataset. For cardinality constraints, our per-iteration privacy parameter ε_0 can be set with either basic composition, advanced composition (e.g., Theorem 5.1) or the analysis for decomposable objective (e.g., Theorem 5.2). We follow [16, 43] and pick the initialization that achieves (ε, δ) -DP with the smallest noise

⁴The implementation can be found at <https://github.com/ronzadi/Differentially-Private-Max-Sum-Diversification>.

scale. For the partition matroid constraint, we pick the best out of basic and advanced composition. We fix $\gamma = 0.1$ for the utility parameter, and by default use $\lambda = 0.5$. Results are averaged over 10 runs.

Baselines. As MSD has not been previously explored in a DP setting, we compare our proposed methods against non-private, random, and DP submodular maximization baselines.

- **Greedy** (Cardinality): The standard non-private greedy algorithm for cardinality constraints [8].
- **LS** (Matroid): The non-private local search algorithm for general matroid constraints [8] discussed in Section 7.
- **Random** (Cardinality/Matroid): Selects a random feasible subset. Since the output distribution of the exponential mechanism converges to uniform as $\varepsilon \rightarrow 0$, this baseline serves as a lower bound on utility for private algorithms. Following prior work [16, 17, 43], we incorporate Random to contextualize the performance of our algorithms.
- **DP-Re1G** (Cardinality / Matroid): The DP greedy algorithm of [43] for DP submodular maximization. It greedily selects feasible elements using the EM, applied only to the submodular component of our objective (i.e., $\lambda = 0$). This baseline isolates the effect of removing the diversity term by optimizing relevance alone.
- **DP-Mix** (Cardinality): A hybrid baseline combining the random baseline with DP-Re1G: $k/2$ items are selected uniformly at random, and the remaining $k/2$ are selected using DP-Re1G.

R1-D1

8.2 Datasets, Objective Functions, and Constraints

We evaluate our approach on two real-world data summarization tasks: Uber location selection and Amazon product selection.

Uber Location Selection. The first application involves selecting a representative summary of Uber pickup locations in Manhattan. This setup adapts the location selection experiments from [16, 43, 44]. The goal is to select a set of locations from a public candidate set, for instance, to serve as waiting spots for idle drivers.

Dataset. We consider a sensitive dataset D of $m = 600,000$ Uber pickup locations in Manhattan from 2014 [28]. Each record consists of longitude and latitude coordinates. For the public ground set, we follow the established methodology and consider a grid of n points over Manhattan, where by default $n = 1000$.

Objective function. For relevance, we adopt the Uber “convenience” score from Mitrovic et al. [44], defined as follows. For a passenger location $a = (x_a, y_a)$ and a grid point $b = (x_b, y_b)$ define $c(a, b) = 2 - \frac{2}{1 + e^{-200\|a-b\|_1}}$, where $\|a-b\|_1 = |x_a - x_b| + |y_a - y_b|$ is the Manhattan distance between the two points. The relevance of a set S is defined as $f_D(S) = \frac{1}{m} \sum_{a \in D} \min_{b \in S} c(a, b)$ with $f_D(\emptyset) = 0$. This function is monotone, submodular, and $1/m$ -decomposable (See Appendix F). However, as observed in [45], it does not inherently promote geographic diversity. For example, suppose congestion slows down traffic in and out of a certain area. If all selected waiting locations are concentrated there, drivers may struggle to reach passengers efficiently. To address this, we add a diversity term defined by the normalized sum of distances between the selected points: $d(S) = \sum_{b, b' \in S} \frac{\|b-b'\|_1}{M}$, where M is an upper bound on the distance within the area such that $\|b-b'\|_1 \in [0, 1]$

Meta

Constraints. We evaluate this task under a cardinality constraint k .

Amazon Product Selection. The second application follows the scenario in Example 1.1 and focuses on selecting a representative summary of highly-purchased Amazon products to maximize user reach.

Dataset. We utilize the *Health and Household* subset of the Amazon Reviews 2023 dataset [37], which contains product metadata (e.g., categories, price) and user review data. We treat the metadata as the public ground set and the review/purchase history as the private dataset. Specifically, we select 3000 products from the *Health Care* category as our ground set V and discretize their prices into four bins. The dataset D contains the purchase records of 1,317,193 users.

Objective function. We consider the objective function from our running example (Example 4.4), where the goal is to select a subset $S \subseteq V$ maximizing user reach, with diversity given by the Jaccard distance over category sets.

Constraints. We consider the intersection of a uniform matroid of rank k with a partition matroid that limits selections to $\lceil k/4 \rceil$ products per price bin. We refer to the latter setting as Amazon-Partition. We also conducted experiments under a cardinality constraint. The scalability trends were similar to those observed for Uber, which is expected since the two settings differ primarily in the cost of evaluating the objective function rather than in the algorithmic behavior. We likewise observed highly similar utility trends, and therefore defer these results to Appendix G.

8.3 Utility Analysis

We examine how the selected subset size k and the privacy parameter ϵ impact the MSD objective value $(1 - \lambda)f_D(S) + \frac{2\lambda}{k(k-1)}d(S)$. By default, we set $k = 20$ for the Uber experiment, $k = 6$ for the Amazon-Partition experiment.

Impact of k . We evaluate the impact of the rank k on the objective value (Figure 7). Overall, our algorithms perform competitively with the non-private baselines and consistently outperform the other DP baselines. For the Uber dataset, our three algorithms remain within 3.4% of the non-private Greedy baseline on average, with the gap increasing in k due to accumulated DP noise. In contrast, DP-Mix is on average 17% below Greedy, DP-ReIG is 20% below, and Random is 26% below. The trends for Amazon-Cardinality are similar and the corresponding plot is omitted. For Amazon-Partition, the utility of DP-SLS remains nearly identical to the non-private LS baseline across the examined range, and is sometimes even slightly better (within 0.1% on average). In contrast, the baselines DP-ReIG and Random perform substantially worse, remaining approximately 25% and 30% below LS, respectively. We also note that the objective value need not increase monotonically with k . This is because the normalization factor of the diversity increases with k , and stricter partition constraints for smaller k may restrict the algorithm to naturally more diverse solutions.

Impact of ϵ . We evaluate the impact of the privacy parameter ϵ on the resulting objective value (Figure 6). As expected, a larger privacy budget improves the utility of all DP algorithms. For the Uber dataset, all our algorithms significantly outperform the DP baselines and achieve utility comparable to the non-private Greedy

baseline, reaching within 1% even at $\epsilon = 0.1$. In contrast, DP-Mix, DP-ReIG, and Random remain substantially lower, at 19%, 27%, and 34% below, respectively. For Amazon-Cardinality, the trends are similar and the corresponding plot is omitted. For Amazon-Partition, we similarly find that at $\epsilon = 0.1$, DP-SLS reaches utility within 1% of the non-private LS baseline and significantly outperforms DP-ReIG and Random, which remain 38% and 48% below LS, respectively. DP-ReIG slightly benefits from lower privacy budgets, as the increased randomness induces more diverse selections, resulting in performance closer to Random in this regime.

Impact of λ . We evaluate the robustness of our algorithms with respect to λ , which controls the trade-off between relevance and diversity. The results, shown in Figure 8, demonstrate that the utility of our algorithms relative to the corresponding non-private baselines remains consistently high across all examined values $\lambda \in [0, 0.8]$, indicating that the observed performance is largely insensitive to the choice of λ . For the Uber dataset, the utility gap from the non-private Greedy baseline remains below 1% on average for DP-Greedy and DP-NOSG, and below 3% for DP-OSG. In contrast, the other DP baselines perform substantially worse, with Random reaching up to 38% lower utility on average. For Amazon-Partition, the utility gap of DP-SLS from the non-private LS baseline remains below a marginal 0.1% across all examined values of λ . Similar trends are observed for Amazon-Cardinality.

As expected, the performance of relevance-only baselines deteriorates as λ increases as diversity becomes more dominant in the objective. When $\lambda = 0$, corresponding to relevance-only optimization, our algorithms achieve essentially the same utility as DP-ReIG. This shows that our approach recovers the performance of prior work in this special case while naturally extending to more general relevance-diversity objectives.

8.4 Performance Analysis

We evaluate the computational efficiency of our algorithms by measuring both query complexity and execution time (Figures 2 and 4). In the submodular maximization literature, the number of oracle calls is considered the most robust metric for performance evaluation, as it is invariant to implementation and problem-specific details [40]. Nevertheless, we also report execution time to provide additional practical insights. Note that we do not attempt to optimize the oracle evaluation time.

Number of Oracle Calls. Figure 2 reports the number of oracle calls as k varies. For cardinality constraints, DP-Greedy matches the non-private Greedy baseline, scaling linearly in k , while DP-NOSG exhibits a logarithmic dependence and significantly reduces query complexity (e.g., 15× fewer calls than Greedy at $k = 100$ on Uber). DP-OSG is even more efficient, with a query count essentially independent of k , achieving up to a 57× reduction at $k = 100$ while maintaining comparable utility. For Amazon-Partition, the query count of the non-private LS baseline is dominated by its initialization phase, which requires an exhaustive $\Theta(n^2)$ search over feasible pairs and is independent of k ; in practice, the subsequent local search terminates after fewer iterations than the worst-case bound. In contrast, DP-SLS avoids this exhaustive initialization and maintains a lower query count.

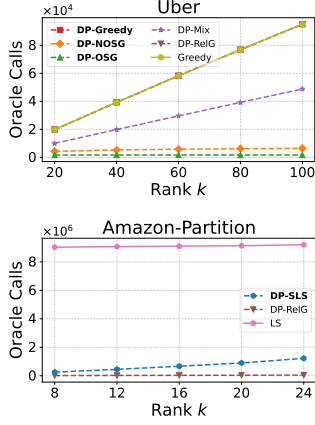


Figure 2: Number of oracle calls as the rank k varies.

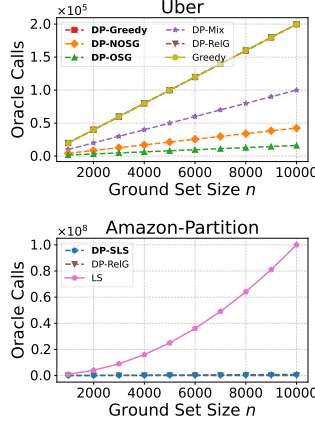


Figure 3: Number of oracle calls as the ground set size n varies.

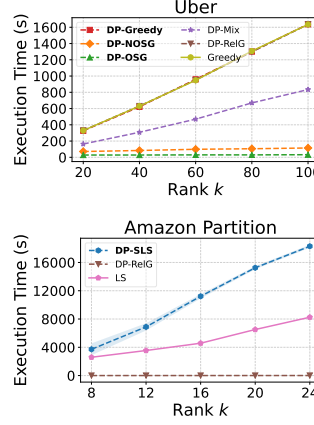


Figure 4: Execution time (seconds) as the rank k varies.

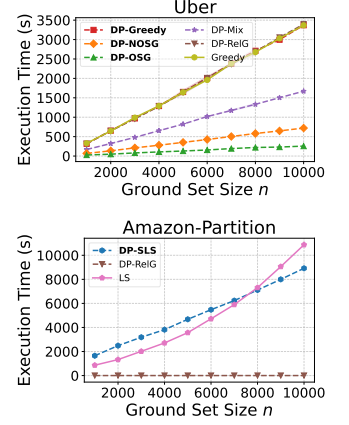


Figure 5: Execution time (seconds) as the ground set size n varies.

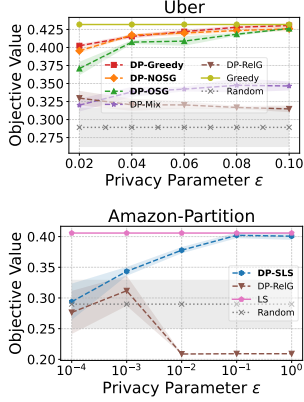


Figure 6: Objective value as the privacy parameter ϵ varies.

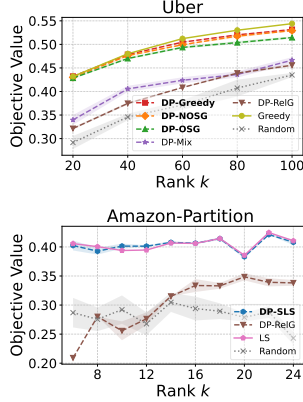


Figure 7: Objective value as the rank (maximal size of a feasible solution) k varies.

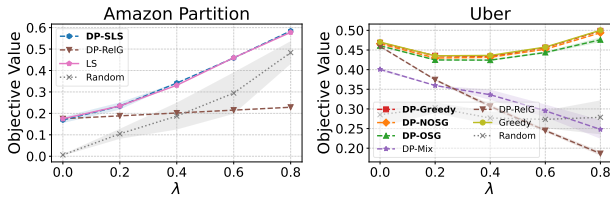


Figure 8: Objective value as the parameter λ varies.

Figure 3 shows the dependence on n . On the Uber dataset, all algorithms exhibit linear scaling in n , while our DP-NOSG and DP-OSG achieve a smaller slope due to their improved dependence on k . For Amazon-Partition, LS exhibits quadratic scaling in n due to its initialization phase, which involves a brute-force search over pairs of elements, whereas our DP-SLS scales linearly.

Execution Time. Figure 4 illustrates execution time as k varies. For cardinality constraints, the trends align with our query complexity analysis. DP-NOSG exhibits a logarithmic dependence on k , outperforming both Greedy and all DP baselines; on Uber ($k = 100$), it is $14\times$ faster than Greedy. DP-OSG is even more efficient, achieving a $50\times$ speedup while preserving comparable utility.

For Amazon-Partition, DP-SLS is slower than the non-private LS baseline despite its fewer oracle calls. This discrepancy arises because the LS bottleneck is dominated by pair evaluations, which are computationally cheaper in this application than evaluating the larger sets required for potential swaps. Moreover, the non-private baseline often converges to a local optimum in substantially fewer iterations than the fixed number required by the DP-SLS theoretical analysis. Overall, DP-SLS is approximately $2\times$ slower than LS on average. While both algorithms may be slow for interactive settings, DP-SLS uniquely provides formal privacy and utility guarantees. In contrast, DP-RelG is significantly faster (minutes rather than hours, and is thus omitted from the plot), but achieves substantially lower utility and lacks a constant-factor approximation guarantee.

Figure 5 depicts scalability as n varies. For cardinality constraints, DP-OSG and DP-NOSG are the most scalable DP methods, and their execution-time trends match the query-complexity trends. For Amazon-Partition, DP-SLS exhibits the improved scalability in n predicted by our bottleneck analysis; for sufficiently large ground sets ($n \geq 8000$), it becomes faster than the LS baseline.

9 CONCLUSION AND LIMITATIONS

In this work, we studied the max-sum diversification problem, a widely adopted formulation introduced by Borodin et al. [8], under differential privacy. We designed DP algorithms for both cardinality and matroid constraints. The proposed algorithms achieve utility guarantees comparable to their non-private counterparts while offering improved complexity, making them of independent interest beyond privacy considerations. We conducted experiments on real-world datasets, showing that the proposed DP algorithms achieve

R2-W3
R1-D2

R2-W3

R2-W3
R1-D2

R2-W4

high utility while substantially improving efficiency in practice for cardinality constraints.

Our work has certain limitations that open interesting directions for future research. First, as stated in Remark 7.3, our current guarantees for matroids apply to arbitrary Δ -sensitive functions; however, they do not yet exploit the decomposability structure to achieve strictly better utility bounds. As noted in that remark, achieving such improvements appears non-trivial. Second, our fastest $O_\gamma(n)$ oracle-call algorithm provides a constant-factor approximation that does not reach the tight $1/2$ ratio. It remains an open question whether a near-optimal $(1/2 - \gamma)$ -approximation can be attained for cardinality constraints with $O_\gamma(n)$ calls. Third, while Algorithm 3 has strictly better worst-case query complexity than the non-private baseline of Borodin et al. [8], it is slower in practice in some regimes due to realizing its worst-case complexity in each execution. A natural question is whether this algorithm can be accelerated in practice while maintaining its high utility. Finally, addressing additional diversity models remains an important direction for future work.

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A SUBMODULAR FUNCTIONS AND MATROIDS

Submodular set functions have numerous applications across areas such as machine learning, databases, economics, and network analysis. Their usefulness stems from the property of diminishing returns, which naturally arises in many practical situations, and informally states that the incremental benefit of adding an element to a smaller set is larger than adding it to a larger set. This property makes submodular functions well-suited to model the notion of relevance.

Definition A.1. [46] A set function $f : 2^V \rightarrow \mathbb{R}$ is submodular if $f(u \mid S) \geq f(u \mid T)$ for every two sets $S \subseteq T \subseteq V$ and element $u \in V \setminus T$.

A set function $f : 2^V \rightarrow \mathbb{R}$ is *non negative* if $f(S) \geq 0$ for every set $S \subseteq V$, and is *monotone* if $f(S) \leq f(T)$ for every two sets $S \subseteq T \subseteq V$. We consider only non-negative, monotone submodular functions, as is the case in the non-private setting [8].

The problem of maximizing a submodular function subject to a *matroid independence* constraint is one of earliest and most studied problems in submodular maximization [53]. In the context of max-sum diversification as well, matroids allow a substantial generalization of the types of constraints that can be modeled compared to simple cardinality [8]. For example, cardinality constraint is a special case called a *uniform matroid*. In a *partition matroid*, the ground set V is partitioned into disjoint subsets P_i , and the independent sets are $\{S \subseteq 2^V : |S \cap P_i| \leq r_i\}$, i.e., sets that satisfy a separate cardinality bound for each block. As another example, a *transversal matroid* models representativeness. Let $A_1, \dots, A_m$ a collection of categories, i.e, subsets of the ground set V (that may overlap). A subset $S \subseteq V$ is independent if there exists an injective mapping $h : S \rightarrow [m]$ such that $u \in A_{h(u)}$ for all $u \in S$, that is, no two items represent to the same category. Additional background an examples can be found in [53]. We now give the formal definition.

Definition A.2. [58] A matroid is a pair $(V, \mathcal{I})$, where $\mathcal{I} \subseteq 2^V$ is a collection of subsets called *independent sets*, with the following properties.

- (i) if $A \subseteq B$ and $B \in \mathcal{I}$ then $A \in \mathcal{I}$ (Hereditary),
- (ii) If $A, B \in \mathcal{I}$ and $|A| < |B|$ then $\exists u \in B \setminus A$ such that $A \cup \{u\} \in \mathcal{I}$ (Exchange).

A matroid is a combinatorial abstraction of the notion of independence from linear algebra. Following the terminology used for linear spaces, independent sets that are inclusion-wise maximal are called *bases*. An immediate corollary of the exchange property is that all bases have the same number of elements, and this number is referred to as the *rank* of the matroid. As is standard in the literature, we assume access to the set functions and the matroid via *oracles*. Specifically, the *value oracle* for f (respectively, d) returns the value $f(S)$ (respectively, $d(S)$) given a set S . The *independence oracle* associated with a matroid $(V, \mathcal{I})$ takes as input a set $S \subseteq V$ and answers whether $S \in \mathcal{I}$. The number of oracle calls performed by an algorithm is commonly used as a proxy for its time complexity, as the exact complexity may be application specific, or depend on implementation details such as the data structures used to maintain sets. In our algorithms, the number of queries to the value and independence oracles are the same up to constant factors. We therefore refer to the total number of queries to both as the number of *oracle calls*.

B OMITTED PROOFS FROM SECTION 5

In this section, we provide the proof of Lemma 5.3, restated next for convenience.

LEMMA 5.3 (ADAPTED FROM [8]). *For all $i \in [k] \triangleq \{1, \dots, k\}$, $d(C_i, S_{i-1}) \geq \frac{(i-1)|C_i|}{k(k-1)} d(\text{OPT})$.*

Let OPT be an optimal solution, i.e., a subset of size at most r that maximizes ϕ . Since ϕ is monotone, we may assume that $|\text{OPT}| = k$. Let S_i be the solution at the end of step i . Define $A_i = S_{i-1} \cap \text{OPT}$, $B_i = S_{i-1} \setminus A_i$ and $C_i = \text{OPT} \setminus A_i$.

LEMMA B.1 ([50]). *Suppose d is a pseudometric, and let X and Y be two disjoint subsets of V . Then, $|Y| \cdot d(X) \leq (|X| - 1) \cdot d(Y, X)$.*

PROOF OF LEMMA B.1. If either set is empty, or if $|X| = 1$ then both sides equal zero and the inequality trivially holds. Thus, we assume that both sets are not empty and that $|X| \geq 2$. For distinct $x, x' \in X$ and for $y \in Y$, we have by the triangle inequality $d(x, x') \leq d(x, y) + d(x', y)$. Summing over all distinct pairs $x, x' \in X$, we obtain $d(X) \leq (|X| - 1)d(y, X)$, because each distance $d(y, x)$ appears in the sum exactly $|X| - 1$ times. Summing over all $y \in Y$, we get $|Y|d(X) \leq (|X| - 1)d(Y, X)$. $\square$

PROOF OF LEMMA 5.3. We first prove the claim for the special case of $C_i = \{o\}$ for some $o \in V \setminus S_{i-1}$, which is not covered by the proof in [8]. Since $|S_{i-1}|$ increases by 1 in each round, it follows that $i = k$. Hence, it remains to prove that $d(o, S_{i-1}) \geq d(\text{OPT})/k$. Note that our assumption implies that $\text{OPT} = S_i \cup \{o\}$. By Lemma B.1, we have

$$d(S_{i-1}) \leq (|S_{i-1}| - 1)d(o, S_{i-1}) \leq (k - 1)d(o, S_{i-1}).$$

Therefore,

$$d(\text{OPT}) = d(S_{i-1} \cup \{o\}) = d(S_{i-1}) + d(o, S_{i-1}) \leq k \cdot d(o, S_{i-1}).$$

For the case $|C_i| > 1$, we follow the proof of Borodin et al. [8]. By Lemma B.1, the following inequalities hold

$$(|C_i| - 1)d(B_i, C_i) \geq |B_i| d(C_i) \quad (6)$$

$$(|C_i| - 1)d(A_i, C_i) \geq |A_i| d(C_i) \quad (7)$$

$$(|A_i| - 1)d(A_i, C_i) \geq |C_i| d(A_i) \quad (8)$$

$$d(A_i, C_i) + d(A_i) + d(C_i) = d(\text{OPT}) \quad (9)$$

where (9) holds since A_i and C_i are disjoint sets whose union equals OPT. Then, we multiply equations (6),(7),(8),(9) by the following non-negative numbers respectively

$$\frac{1}{|C_i| - 1}, \frac{|C_i| - |B_i|}{k(|C_i| - 1)}, \frac{i - 1}{k(k - 1)}, \frac{(i - 1)|C_i|}{k(k - 1)}.$$

Summing the multiplied equations, we get

$$d(A_i, C_i) + d(B_i, C_i) - \frac{(i - 1)|C_i|(k - |C_i|)}{k(k - 1)(|C_i| - 1)}d(C_i) \geq \frac{(i - 1)|C_i|}{k(k - 1)}d(\text{OPT}).$$

Since $|C_i| \leq k$ and $d(A_i, C_i) + d(B_i, C_i) = d(C_i, S_{i-1})$, we have

$$d(C_i, S_{i-1}) \geq \frac{(i - 1)|C_i|}{k(k - 1)}d(\text{OPT}),$$

and the proof is complete. $\square$

C OMITTED PROOFS FROM SECTION 6

The proof of the query complexity of Algorithm 2 is given by the following lemma.

LEMMA C.1. *Algorithm 2 makes $O(n \log k \log(1/\gamma))$ queries when $g(i) = k - i + 1$ for all $i \in [k]$, and $O(n \log(1/\gamma))$ queries when $g(i) = \min\{k, n - i + 1\}$ for all $i \in [k]$.*

PROOF. The number of queries performed by Algorithm 2 in iteration i is $|V_i| + 1$, and we note that

$$|V_i| \leq \frac{|N_i| \log(1/\gamma)}{g(i)} + 1.$$

Since a new element is added to S_i in each iteration, we have $|N_i| = |V \setminus S_{i-1}| = n - i + 1$. Hence, the total number of queries is bounded by

$$\sum_{i=1}^k |V_i| + 1 \leq 2k + \log(1/\gamma) \sum_{i=1}^r \frac{n - i + 1}{g(i)}.$$

If $g(i) = r - i + 1$, the standard harmonic sum bound yields $\sum_{i=1}^r \frac{n - i + 1}{g(i)} = O(n \log k)$, and the first part of the lemma follows. If $g(i) = \min\{k, n - i + 1\}$, we have

$$\sum_{i=1}^r \frac{n - i + 1}{g(i)} \leq \sum_{i=1}^r \frac{n - i + 1}{r} + \sum_{i=1}^r \frac{n - i + 1}{n - i + 1} \leq n + r = O(n),$$

which implies the second part. $\square$

We now turn to complete the proof for the privacy guarantee stated in Theorem 6.3. To this end, we utilize the following concentration bound.

LEMMA C.2 (CLAIM 3.8 IN [16], BASED ON [33]). *Consider an n -round probabilistic process. In each round $i \in [n]$, an adversary chooses a distribution $\mathcal{P}_i$ over $[0, 1]$ and a sample R_i is drawn from this distribution. Let $Z_1 = 1$ and $Z_{i+1} = Z_i - R_i Z_i$. We define the random variable $Y_j = \sum_{i=j}^n Z_i \mathbb{E}[R_i]$. Then for any $j \in [n]$, $\Pr[Y_j \geq qZ_j] \leq \exp(3 - q)$.*

The above bound and the argument of [16, 33] yields the following lemma. We continue with the notation established in the proof of Theorem 6.3.

LEMMA C.3. *The following inequality holds.*

$$\Pr \left[\prod_{i=1}^k \mathbb{E}_{u \sim \mathcal{P}_i} [\exp(\frac{\epsilon_0}{2} \cdot \beta_i^i(u))] \geq (e^{\epsilon_0/2} - 1)(3 + \log(1/\delta)) \right] \leq \delta$$

PROOF. For all $\varepsilon_0 \in (0, 1]$, we have

$$\begin{aligned} \prod_{i=1}^k \mathbb{E}_{u \sim \mathcal{P}_i} [\exp(\frac{\varepsilon_0}{2} \cdot \beta_t^i(u))] &\leq \prod_{i=1}^k \mathbb{E}_{u \sim \mathcal{P}_i} [1 + (e^{\varepsilon_0/2} - 1) \cdot \beta_t^i(u)] \\ &\leq \exp\left((e^{\varepsilon_0/2} - 1) \sum_{i=1}^k \mathbb{E}_{u \sim \mathcal{P}_i} [\beta_t^i(u)]\right). \end{aligned}$$

where the first inequality holds since $e^x \leq 1 + \frac{e^{\varepsilon_0/2} - 1}{\varepsilon_0/2} x$ for all $x \in [0, \varepsilon_0/2]$, and the second uses $1 + x \leq e^x$ for all x . It remains to show that the sum of expectations is bounded with high probability. Consider the following probabilistic process. In each round $i \in [k]$, u_i is drawn from $\mathcal{P}_i$. Let $Z_i = 1 - \phi'_t(U_{i-1})$ be the total remaining marginal utility at the beginning of iteration i . Define the random variable $R_{i-1}(u) = \beta_t^i(u)/Z_{i-1}$ which is determined by the sample of $u \sim \mathcal{P}_i$, and denotes the percentage increase in utility in iteration i (If $Z_i = 0$, we set $R_i = 0$). Since ϕ'_t is monotone and has range in $[0, 1]$, we have $R_{i-1}(u) \in [0, 1]$. Note that using these definitions, we have $Z_i = Z_{i-1} - R_{i-1}Z_{i-1}$. Define, $Y_j = \sum_{i=j}^k Z_i \cdot \mathbb{E}_{u \sim \mathcal{P}_i} [R_i(u)]$, and observe that the sum of interest is exactly Y_1 . Letting $q = 3 + \log(1/\delta)$ and using Lemma C.2, we conclude that

$$\Pr\left[\sum_{i=1}^k \mathbb{E}_{u \sim \mathcal{P}_i} [\beta_t^i(u)] \geq q\right] = \Pr[Y_1 \geq q] \leq \Pr[Y_1 \geq qZ_1] \leq \exp(3 - q) = \delta$$

where the second inequality holds since $Z_1 \in [0, 1]$. $\square$

The remainder of this section is devoted to the proof of Theorem 6.5, restated next.

THEOREM 6.5. *Suppose ϕ has sensitivity Δ . Then, DP-OSG (Algorithm 2) outputs a feasible set S such that*

$$\mathbb{E}[\phi(S)] \geq (1 - (\frac{2}{e})^{(1-\gamma)(1-1/k)}) \cdot \phi(\text{OPT}) - O(\frac{k\Delta \log n}{\varepsilon_0})$$

and makes $O(n \log \gamma^{-1})$ oracle calls.

LEMMA 6.7. *For every iteration $i \geq 2$,*

$$d(\text{OPT} \cup S_{i-1}) - d(S_{i-1}) \leq \left(1 + \frac{k-1}{i-1}\right) \cdot d(C_i, S_{i-1}).$$

PROOF OF LEMMA 6.7. By Lemma B.1, we have $|S_{i-1}| \cdot d(C_i) \leq (|C_i| - 1) \cdot d(C_i, S_{i-1})$. Moreover, since $|S_{i-1}| = i - 1$ and $|C_i| \leq k$, we have $d(C_i) \leq \frac{k-1}{i-1} \cdot d(C_i, S_{i-1})$. Therefore,

$$d(\text{OPT} \cup S_{i-1}) - d(S_{i-1}) = d(C_i \cup S_{i-1}) - d(S_{i-1}) = d(C_i) + d(C_i, S_{i-1}) \leq \left(1 + \frac{k-1}{i-1}\right) d(C_i, S_{i-1}).$$

where the second equality holds because S_{i-1} and C_i are disjoint. $\square$

LEMMA C.4. *The following inequalities hold:*

$$\prod_{i=1}^{k-1} \left(1 - \frac{1-\gamma}{k(1 + \frac{k-1}{i})}\right) \leq \left(\frac{2}{e}\right)^{(1-\gamma)(1-1/k)} \leq \frac{2}{e} + \gamma + \frac{1}{k}.$$

PROOF. We first evaluate the sum within the exponent. We have:

$$\begin{aligned} \frac{1}{k} \sum_{i=1}^{k-1} \frac{i}{i+k-1} &= \frac{1}{k} \sum_{i=1}^{k-1} \left(1 - \frac{k-1}{i+k-1}\right) = \frac{k-1}{k} - \frac{k-1}{k} \cdot \sum_{i=1}^{k-1} \frac{1}{i+k-1} \\ &= \frac{k-1}{k} - \frac{k-1}{k} \cdot \sum_{j=k}^{2k-2} \frac{1}{j} \geq \frac{k-1}{k} - \frac{k-1}{k} \cdot \log(2) \\ &= \left(1 - \frac{1}{k}\right)(1 - \log 2), \end{aligned}$$

where in the inequality we have used the integral bound:

$$\sum_{j=k}^{2k-2} \frac{1}{j} \leq \int_{k-1}^{2k-2} \frac{1}{x} dx = \log(2k-2) - \log(k-1) = \log(2).$$

We next observe that the bound $\log(1-x) \leq -x$, which holds for all $x \in (0, 1)$, together with the preceding derivation, implies that:

$$\sum_{i=1}^{k-1} \log \left(1 - \frac{1-\gamma}{k(1+\frac{k-1}{i})} \right) \leq -(1-\gamma) \sum_{i=1}^{k-1} \frac{1}{k(1+\frac{k-1}{i})} \leq -(1-\gamma) \left(1 - \frac{1}{k} \right) (1 - \log 2).$$

The first inequality in the lemma follows by exponentiating both sides of the result above. For the second inequality, observe that:

$$\begin{aligned} (2/e)^{(1-\gamma)(1-1/k)} &\leq (2/e)^{(1-\gamma-1/k)} = e^{(1-\gamma-1/k) \log(2/e)} = \frac{2}{e} \cdot e^{\log(e/2)(\gamma+1/k)} \\ &\leq \frac{2}{e} \cdot (1 + (e-1) \log(e/2)(\gamma+1/k)) \leq \frac{2}{e} + \gamma + \frac{1}{k}, \end{aligned}$$

where the second inequality follows from the fact that $e^x \leq 1 + (e-1)x$ for all $x \in [0, 1]$. $\square$

D OMITTED PROOFS FROM SECTION 7

The following lemma is a known property of matroids.

LEMMA D.1 ([9]). *Let X and Y be two bases of a matroid $\mathcal{M} = (V, \mathcal{I})$. Then, there exists a bijection $h : X \rightarrow Y$ such that for every $u \in X$, $Y - h(u) + u \in \mathcal{I}$, and $h(u) = u$ for every $u \in X \cap Y$.*

We use the following notation, some of which has been introduced in Section 7. Let S_i denote the base selected at iteration i , $C_i = \text{OPT} \setminus S_{i-1}$, $B_i = S_{i-1} \setminus \text{OPT}$, and let $h : S_{i-1} \rightarrow \text{OPT}$ be a bijection satisfying the condition in Lemma D.1. Denote $B_i = \{b_1, \dots, b_t\}$, and let $c_j = h(b_j)$ for each $j \in [t]$. The following result was proved in [8].

LEMMA D.2 (LEMMA 5 AND 7, [8]). *Suppose $\mathcal{M}$ has rank $r > 2$ and $t \geq 2$. Then,*

- (i) $\sum_{j=1}^t [f(S_{i-1} - b_j + c_j) - f(S_{i-1})] \geq f(\text{OPT}) - 2f(S_{i-1})$,
- (ii) $\sum_{j=1}^t [d(S_{i-1} - b_j + c_j) - d(S_{i-1})] \geq d(\text{OPT}) - 2d(S_{i-1})$.

PROOF OF LEMMA 7.4. Recall that $h(u) = u$ for every $u \in S_{i-1} \cap \text{OPT}$. Thus, when $t = 0$, we have $S_{i-1} = \text{OPT}$, so the inequality becomes $r\phi(\text{OPT}) \geq (r-1)\phi(\text{OPT})$. When $t = 1$, the left-hand side becomes $\phi(\text{OPT}) + (r-1)\phi(S_{i-1})$. Since the claim holds in both cases, we assume $t \geq 2$. We get

$$\begin{aligned} \sum_{u \in S_{i-1}} \phi(S_{i-1} - u + h(u)) &= \sum_{i=1}^t \phi(S_{i-1} - b_i + c_i) + (r-t)\phi(S_{i-1}) \\ &\geq \phi(\text{OPT}) + (t-2)\phi(S_{i-1}) + (r-t)\phi(S_{i-1}) \\ &= \phi(\text{OPT}) + (r-2)\phi(S_{i-1}). \end{aligned}$$

where the inequality follows by summing the inequalities in Lemma D.2 and rearranging. $\square$

LEMMA D.3. *Let $V_i \subseteq V$ be a uniformly random subset of size $\lceil \frac{n}{r} \rceil$. Then,*

$$\Pr[(S_{i-1} \times V_i) \cap M_i \neq \emptyset] \geq 1 - 1/e.$$

PROOF OF LEMMA D.3. Consider the probability that V_i does not contain any element of $\{h(u) : u \in S_{i-1}\}$. If $|V_i| > n - r$, this probability is zero. Otherwise, it equals

$$\frac{\binom{n-k}{|V_i|}}{\binom{n}{|V_i|}} = \prod_{i=0}^{|V_i|-1} \frac{n-k-i}{n-i} = \prod_{i=1}^{|V_i|} \left(1 - \frac{r}{n-i+1} \right) \leq \left(1 - \frac{r}{n} \right)^{|V_i|} \leq e^{-\frac{r|V_i|}{n}} \leq e^{-1}.$$

where the last inequality holds since $r \cdot |V_i| = r \cdot \lceil \frac{n}{r} \rceil \geq n$. The claim follows. $\square$

PROOF OF LEMMA 7.5. Denote $\alpha = 4\Delta \log(n)/\epsilon_0$. Fix an iteration $i \in [T]$ and let us condition on having selected a base S_{i-1} after iteration $i-1$, and on some realization of the set V_i at iteration i . Let $M_i = \{(u, h(u)) : u \in S_{i-1}\}$. By Theorem 3.5, the exponential mechanism guarantees that

$$\begin{aligned} \mathbb{E} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] &\geq \max_{(u,v) \in W_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] - \alpha \\ &= \max \left\{ 0, \max_{\substack{(u,v) \in S_{i-1} \times (V_i \setminus S_{i-1}) \\ S_{i-1} - u + v \in \mathcal{I}}} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] \right\} - \alpha \\ &\geq \max \left\{ 0, \max_{(u,v) \in (S_{i-1} \times V_i) \cap M_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] \right\} - \alpha \\ &\geq \max \left\{ 0, \sum_{(u,v) \in (S_{i-1} \times V_i) \cap M_i} \frac{\phi(S_{i-1} - u + v) - \phi(S_{i-1})}{|(S_{i-1} \times V_i) \cap M_i|} \right\} - \alpha \end{aligned}$$

where the equality in the second line holds since W_i always contains a dummy swap that does not change the current solution. Unfixing the implicit conditioning on V_i in iteration i and taking expectation over all their possible realizations, we get

$$\begin{aligned}\mathbb{E}[\phi(S_{i-1} - u + v) - \phi(S_{i-1})] &\geq \mathbb{E}_{V_i} \left[\max \left\{ 0, \sum_{(u,v) \in (S_{i-1} \times V_i) \cap M_i} \frac{\phi(S_{i-1} - u + v) - \phi(S_{i-1})}{|(S_{i-1} \times V_i) \cap M_i|} \right\} - \alpha \right] \\ &\geq \max \left\{ 0, \mathbb{E}_{V_i} \left[\sum_{(u,v) \in (S_{i-1} \times V_i) \cap M_i} \frac{\phi(S_{i-1} - u + v) - \phi(S_{i-1})}{|(S_{i-1} \times V_i) \cap M_i|} \right] \right\} - \alpha \\ &\geq \frac{1 - 1/e}{r} \sum_{(u,v) \in M_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] - \alpha.\end{aligned}\tag{10}$$

The second inequality follows from Jensen's inequality, as the function $f(x) = \max\{0, x\}$ is convex. We now focus on proving the last inequality. First, if the sum in (10) is negative, the final inequality holds trivially; we therefore assume without loss of generality that it is non-negative. Consider the following probabilistic experiment. We sample V_i as in Algorithm 3. If $(S_{i-1} \times V_i) \cap M_i \neq \emptyset$, we select a uniformly random $(u, v) \in (S_{i-1} \times V_i) \cap M_i$. Otherwise, output $\perp$. Let $p_{u,v}$ denote the probability that $(u, v) \in M_i$ is selected conditioned on the event $(S_{i-1} \times V_i) \cap M_i \neq \emptyset$, which we denote by G . We have

$$\begin{aligned}\mathbb{E}_{V_i} \left[\sum_{(u,v) \in (S_{i-1} \times V_i) \cap M_i} \frac{\phi(S_{i-1} - u + v) - \phi(S_{i-1})}{|(S_{i-1} \times V_i) \cap M_i|} \right] &= \Pr[G] \cdot \sum_{(u,v) \in M_i} p_{u,v} \cdot [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] \\ &\geq \frac{1 - 1/e}{r} \sum_{(u,v) \in M_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})]\end{aligned}$$

where in the first equality we use the law of total expectation, observing that the inner sum is empty and thus equals 0 conditioned on $\bar{G}$. In the second inequality, we observe that since V_i is chosen uniformly at random, the probabilities $p_{u,v}$ must be the same for all $(u, v) \in M_i$, and therefore equal $1/|M_i| = 1/r$. By Lemma D.3 we have $\Pr[G] \geq 1 - 1/e$, and the inequality (10) follows. Using Lemma 7.4, we get

$$\mathbb{E}[\phi(S_{i-1} - u + v)] \geq \phi(S_{i-1}) + \frac{1 - 1/e}{r} (\phi(\text{OPT}) - 2\phi(S_{i-1})) - \alpha.$$

Finally, the lemma follows by unfixing S_{i-1} and taking an expectation over its possible realizations. $\square$

E LOCAL SEARCH FOR RANK 2 MATROID

We follow the notation of Section 7. The difficulty arises because the second inequality in Lemma D.2 may fail when $k = 2$. Nevertheless, as its proof shows, Lemma 7.4 holds for $t = 0$ or $t = 1$ even with $k = 2$, so Lemma 7.5 remains valid in these cases. Therefore, to complete the proof of Theorem 7.1, it suffices to verify that the inequality in Lemma 7.5 continues to hold in iterations with $k = t = 2$, namely when $S_{i-1} = \{b_1, b_2\}$ and $C_i = \{c_1, c_2\}$.

LEMMA E.1. *Suppose that $\mathcal{M}$ has rank $r = 2$. Conditioned on having selected a base $S_{i-1} = \{b_1, b_2\}$ after iteration $i - 1$, the following inequality holds.*

$$\mathbb{E}[\phi(S_i)] \geq \phi(S_{i-1}) + \frac{1 - 1/e}{r} (\phi(\text{OPT}) - 2\phi(S_{i-1})) - \frac{4\Delta \log(n)}{\varepsilon_0}$$

PROOF. Since $\mathcal{M}$ has rank 2, our assumption implies that $\text{OPT} = \{c_1, c_2\}$. Without loss of generality, assume that $f(b_1) \geq f(b_2)$. By the triangle inequality,

$$d(S_{i-1} - b_2 + c_1) + d(S_{i-1} - b_2 + c_2) = d(b_1, c_1) + d(b_1, c_2) \geq d(c_1, c_2) = d(\text{OPT}).$$

Moreover,

$$\begin{aligned}f(\text{OPT}) - f(S_{i-1}) &\leq f(\{b_1, b_2, c_1, c_2\}) - f(\{b_1, b_2\}) \\ &\leq f(\{b_1, b_2, c_1\}) + f(\{b_1, b_2, c_2\}) - 2 \cdot f(\{b_1, b_2\}) \\ &\leq f(\{b_1, c_1\}) + f(\{b_1, c_2\}) - 2 \cdot f(\{b_1\}) \\ &= f(S_{i-1} - b_2 + c_1) + f(S_{i-1} - b_2 + c_2) - 2 \cdot f(\{b_1\})\end{aligned}$$

where the first inequality follows from monotonicity, and the second and third follow from submodularity. Rearranging and using the fact that $f(S_{i-1}) \leq f(\{b_1\}) + f(\{b_2\}) \leq 2f(\{b_1\})$ yields

$$f(S_{i-1} - b_2 + c_1) + f(S_{i-1} - b_2 + c_2) \geq f(\text{OPT}).$$

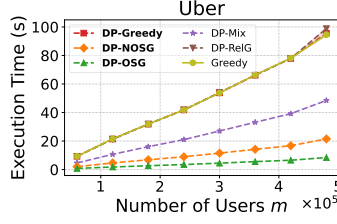


Figure 9: Execution time (seconds) as the number of users m varies.

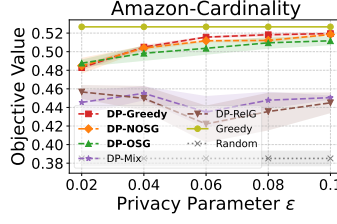


Figure 10: Objective value as the privacy parameter ϵ varies.

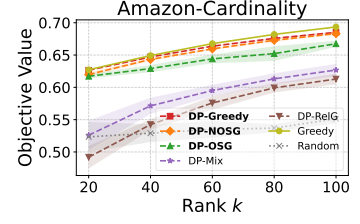


Figure 11: Objective value as the rank k varies.

Now, let $M_i = \{(b_2, c_1), (b_2, c_2)\}$. By summing the previous inequalities, we obtain

$$\sum_{(u,v) \in M_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] \geq \phi(\text{OPT}) - 2\phi(S_{i-1}).$$

The remainder of the proof is analogous to that of Lemma 7.5, but uses the current definition of M_i . A similar derivation yields

$$\begin{aligned} \mathbb{E} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] &\geq \frac{1-1/e}{r} \sum_{(u,v) \in M_i} [\phi(S_{i-1} - u + v) - \phi(S_{i-1})] - \alpha \\ &\geq \frac{1-1/e}{r} (\phi(\text{OPT}) - 2\phi(S_{i-1})) - \alpha \end{aligned}$$

using that $\Pr[\{c_1, c_2\} \cap V_i \neq \emptyset] \geq 1 - e^{-1}$, by an argument similar to that used in the proof of Lemma D.3. $\square$

F SENSITIVITY ANALYSIS FOR THE UBER OBJECTIVE

In this section, we show that the Uber objective defined in Section 8 is $1/m$ -decomposable. The derivation follows a similar logic to that of the Amazon product selection objective presented in Example 4.4.

For a customer location $a = (x_a, y_a)$ and a grid point $b = (x_b, y_b)$, the convenience score $c(a, b) \in [0, 1]$ is defined as in Section 8. The relevance of a set S of selected grid points is given by $f_D(S) = \frac{1}{m} \sum_{a \in D} \max_{b \in S} c(a, b)$. Furthermore, the dispersion $d(S)$ depends only on the distance $d(b, b')$ between selected grid points, which is independent of the sensitive data and is also normalized to $[0, 1]$.

To establish decomposability, we associate each pickup location $a \in D$ with a monotone submodular utility function $f_a(S) = \max_{b \in S} c(a, b)$. This function represents the convenience score of the set S for individual a and depends exclusively on that specific individual. For $\lambda \in [0, 1]$, we define the per-user objective as follows:

$$\phi_a(S) = \frac{(1-\lambda)}{m} f_a(S) + \frac{2\lambda}{mk(k-1)} d(S)$$

where $m = |D|$. The normalization ensures that $\phi_a(S) \in [0, 1/m]$. The total objective is the sum of these individual contributions:

$$\phi(S) = \sum_{a \in D} \phi_a(S) = \frac{1-\lambda}{m} \sum_{a \in D} f_a(S) + \frac{2\lambda}{k(k-1)} d(S).$$

Consequently, ϕ is $1/m$ -decomposable. To derive the sensitivity bound for ϕ , we apply the same argument as in Example 4.4. If $D = (D' \setminus \{x'\}) \cup \{x\}$, then for any $S \subseteq V$, $|\phi_D(S) - \phi_{D'}(S)| = |\phi_x(S) - \phi_{x'}(S)| \leq 1/m$, and thus ϕ has sensitivity bounded by $1/m$.

G ADDITIONAL EXPERIMENTAL RESULTS

In this section, we provide supplementary experimental evaluations to further validate the robustness and scalability of our proposed algorithms.

Execution time by m . Figure 9 shows the scaling of execution time with respect to the number of users $m = |D|$ for the Uber location selection experiment. We emphasize that $|D|$ does not directly affect the algorithmic complexity. Rather, its impact is application-specific, as it only influences the cost of evaluating the objective oracle, which may vary significantly across different instantiations of our framework. Nevertheless, we include these results for completeness.

The results demonstrate an essentially linear dependence on m , with a slightly sharper increase only at the largest scale. Notably, our accelerated algorithms, DP-NOSG and DP-OSG, exhibit smaller constant factors than the baselines, further validating the practical efficiency and scalability of our approach.

Amazon-Cardinality Utility. Figure 10 shows the impact of the privacy parameter ϵ on utility for the Amazon product summarization dataset under a cardinality constraint with $k = 20$, while Figure 11 illustrates the effect of varying k for $\epsilon = 0.1$. Overall, the results show that our algorithms consistently outperform the DP baselines and achieve utility competitive with the non-private Greedy baseline. Performance experiments are omitted, as the observed trends are similar to those in the Uber experiment.